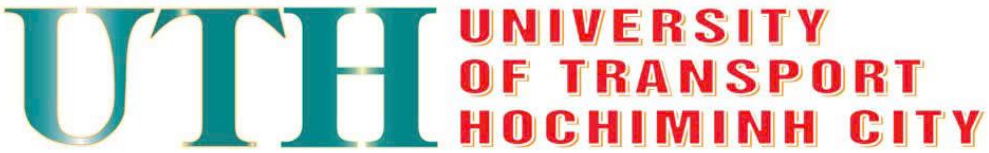
UNIVERSITY OF TRANSPORT HO CHI MINH CITY

INSTITUTE OF INFORMATION TECHNOLOGY



COURSE PROJECT

SOFTWARE ENGINEERING

HOTEL ROOM BOOKING SYSTEM

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Once again, with heartfelt gratitude, our group wishes her good health, happiness, and continued success in her teaching and research.

Respectfully yours,

CHAPTER 1. FUNCTIONAL REQUIREMENTS

1.1 Requirements Engineering Process Summary

Table 1.1: Requirements Engineering Process Summary

Activity	Description and Deliverables
1. Elicitation and Analysis	Identify stakeholder needs (Guest, Customer, Receptionist, Admin) via domain analysis and survey of existing hotel systems.
2. Specification	Document functional requirements using standard templates (ID, Actor, Acceptance Criteria, BRs, Error Handling).
3. Validation	Peer-review requirements for Completeness, Consistency, Verifiability, Traceability, and Feasibility.
4. Change Management	Baseline approved requirements, changes follow formal impact analysis via the Traceability Matrix.

1.2 Functional Requirements Index (FR-01 to FR-30)

Table 1.2: Functional Requirements Index

ID	Group	Function Name	Actor	Pri	UC	Screen
FR-01	Accounts	Register customer account	Guest	M	UC-01	SCR-02
FR-02	Accounts	Log in and log out	All Roles	M	UC-02	SCR-01
FR-03	Accounts	Reset forgotten password	All Roles	S	UC-14	SCR-03
FR-04	Accounts	View and update profile	Customer	S	UC-13	SCR-04
FR-05	Accounts	Manage staff accounts	Admin	M	UC-17	SCR-20
FR-06	Room Search	Search available rooms	Guest, Customer	M	UC-03	SCR-05/06
FR-07	Room Search	Filter and sort results	Guest, Customer	S	UC-03	SCR-06

ID	Group	Function Name	Actor	Pri	UC	Screen
FR-08	Room Search	View room details	Guest, Customer	M	UC-03	SCR-07
FR-09	Booking	Create booking	Customer	M	UC-04	SCR-08
FR-10	Booking	Check overlap and hold room (24h)	System	M	UC-04	SCR-08
FR-11	Booking	Modify booking	Customer	S	UC-04	SCR-12
FR-12	Booking	Cancel booking	Customer, Receptionist	M	UC-05	SCR-12/14
FR-13	Booking	View booking history and status	Customer	S	UC-04	SCR-11
FR-14	Booking	Confirm or reject booking	Receptionist	M	UC-06	SCR-14
FR-15	Booking	Create walk-in booking	Receptionist	S	UC-16	SCR-15
FR-16	Check-in/out	Check in guest	Receptionist	M	UC-07	SCR-16
FR-17	Check-in/out	Assign or change room	Receptionist	S	UC-07	SCR-16
FR-18	Check-in/out	Record extra services	Receptionist	C	UC-08	SCR-17
FR-19	Check-in/out	Check out guest and compute bill	Receptionist	M	UC-08	SCR-18
FR-20	Payment	Pay deposit or full amount (simulated)	Customer, Receptionist	M	UC-09	SCR-09
FR-21	Payment	Create and print invoice	Receptionist	M	UC-09	SCR-18
FR-22	Payment	Refund on cancellation	Receptionist	S	UC-05	SCR-14
FR-23	Room Mgmt	Manage room types	Admin	M	UC-10	SCR-21

ID	Group	Function Name	Actor	Pri	UC	Screen
FR-24	Room Mgmt	Manage rooms and room status	Admin, Receptionist	M	UC-10	SCR-22
FR-25	Room Mgmt	Manage price list	Admin	M	UC-11	SCR-23
FR-26	Room Mgmt	Manage add-on services	Admin	C	UC-10	SCR-24
FR-27	Reports	View revenue report	Admin	S	UC-12	SCR-25
FR-28	Reports	View occupancy report	Admin	S	UC-12	SCR-26
FR-29	Reports	Export report to PDF/Excel	Admin	C	UC-12	SCR-26
FR-30	Notifications	Send booking confirmation	System	S	UC-15	SCR-10

1.3 Business Rules (BR-01 to BR-35) and Authorization

Table 1.3:Business Rules Summary

Group	Domain	Specification Summary
BR-01..05	Accounts and Access	Unique email logins, ≥8-char passwords (hashed). 5 failed logins lock account. Admin manages staff.
BR-06..08	Search and Validation	Stays ≤ 30 nights. Availability requires no overlapping Pending/Confirmed/Checked-in bookings and non-maintenance status.
BR-09..12	Booking Life Cycle	Pending → Confirmed → Checked-in → Completed. Pending holds room 24h. Cancellation refund: 100% (≥48h), 50% (24-48h), 0% (<24h).
BR-13..17	Deposit and Codes	Deposit = 30% of room charge (rounded to 1,000 VND). Booking code: HB-YYYYMMDD-NNNN. Walk-ins require no account.
BR-18..22	Front Desk	Check-in requires Confirmed status. Late check-out (>12:00) adds 50% night charge. Completed check-out sets room to Cleaning.

Group	Domain	Specification Summary
BR-23..27	Payments and Invoices	Methods: Cash, Simulated Card/E-Wallet. Invoice: INV-YYYYMMDD-NNNN (immutable). Check-out completes at 0 balance.
BR-28..32	Inventory and Rates	Unique room numbers. Deletion blocked if active/future bookings exist. Pricing defined per date period without overlaps.
BR-33..35	Reports and Emails	Revenue = Payments - Refunds. Occupancy = Occupied room-nights / (Rooms in service * Days) * 100%. Confirmation emails retried up to 3 times.

Common Requirements (CR-01..10) and System Messages (MSG-01..61):

30-min session timeout (CR-01), dd/mm/yyyy and VND currency (CR-02), Dual validation (CR-03), RBAC enforced on server (CR-04), Audit log for staff mutations (CR-05), Paged lists at 10 rows/page (CR-07), Passwords masked (CR-08). System uses 61 standard messages (MSG-01 to MSG-61) for standardized UX

CHAPTER 2. NON-FUNCTIONAL REQUIREMENTS

2.1 External interfaces

The system communicates with the outside through four groups of interfaces:

Table 2.1: External Interfaces

Type	Interface	Description	Related
User	Screens SCR-01 to SCR-26	Web or desktop interface with role-based access for 4 roles (Guest, Customer, Receptionist, Admin).	CR-02, CR-09, Chapter 7
Software	Database	Relational database (MySQL or SQL Server) storing accounts, rooms, bookings and payments.	Chapter 6, 5.6.4
Software	Email service (external)	Sends booking confirmations and password reset codes by email.	FR-03, FR-30, UC-15
Software	Payment gateway (simulated)	Mock gateway that handles payment requests and returns success or failure.	FR-20, BR-23, UC-09
Hardware and files	Printer, PDF and Excel	Prints invoices and saves them as PDF; exports reports to PDF or Excel.	FR-21, FR-29
Communication	Network (SMTP/HTTP)	Network connection to the database and the mail server through standard protocols.	MSG-58, CR-09

2.2 Quality attributes

2.2.1 Usability

NFR-01 (Should): The system should let a new customer with an account complete a booking, from the home page to the confirmation screen, within 5 minutes without help.

- Measure: test with 5 new users; at least 4 of 5 finish within the time.

Related: FR-06 to FR-09, SCR-05 to SCR-10.

2.2.2 Security

NFR-02 (Must): The system shall reject on the server every request for a function beyond the user's role in the authorization matrix (1.5).

- Measure: 100% of unauthorized requests are rejected with MSG-56. Related: CR-04, BR-05, BR-14.

NFR-03 (Must): The system shall never store, display or log a user password in plain text.

- Measure: 100% of passwords are hashed in the database, and password fields are masked on screen. Related: BR-02, CR-08.

NFR-04 (Should): The system should show personal data (ID number, phone number, email) only to authorized roles.

- Measure: the ID number is visible only to the Receptionist, and a customer cannot view another customer's data. Related: BR-14, 1.5, FR-16.

2.2.3 Maintainability

NFR-05 (Should): The system should keep business-rule parameters (deposit rate, room hold time, VAT) in a configuration file instead of hard-coding them.

- Measure: changing a parameter in the configuration needs no recompilation of the source code. Related: BR-10 to BR-13.

2.2.4 Performance

NFR-06 (Must): The system shall return room search results within 3 seconds and answer other requests within 5 seconds, under a load of 50 concurrent users.

- Measure: load test; at least 95% of requests meet the time limit. Related: FR-06, FR-27 to FR-29.

2.2.5 Reliability

NFR-07 (Must): The system shall not lose data or leave a booking or payment half saved when the application or the database fails suddenly.

- Measure: crash test repeated 10 times, 0% of records in an error state. Related: FR-09, FR-10, FR-20, FR-22.

2.2.6 Compatibility

NFR-08 (Must): The application shall run stably and give the same results on Windows 10 and Windows 11.

- Measure: the 12 main use cases run successfully, and PDF and Excel files open without errors. Related: FR-21, FR-29.

2.3 NFR classification

The NFRs fall into 3 classes by origin:

Table 2.2: NFR Classification

Class	NFR	Reason
Product	NFR-01, NFR-02, NFR-03, NFR-06, NFR-07	Define the behavior of the running system (speed, security, ease of use).
Organizational	NFR-05, NFR-08	Follow the coding standard of the team and the computer environment of the hotel.
External	NFR-04	Comply with the Law on Personal Data Protection 2025 and Decree 356/2025/NĐ-CP.

2.4 NFR table (NFR-xx)

Index of IDs, attributes and MoSCoW priority (M = Must, S = Should):

Table 2.3: Non-Functional Requirements (NFR-01 to NFR-08)

ID	Attribute	Requirement	Priority	Class	Related
NFR-01	Usability	Book within 5 minutes without help	S	Product	FR-06 to FR-09
NFR-02	Security	Server rejects every action outside the user's role	M	Product	CR-04, 1.5

ID	Attribute	Requirement	Priority	Class	Related
NFR-03	Security	Passwords hashed, never shown in plain text	M	Product	BR-02, CR-08
NFR-04	Security	Personal data (ID number, phone) visible to authorized roles only	S	External	BR-14, FR-16
NFR-05	Maintainability	Business parameters kept in a configuration file	S	Organizational	BR-10 to BR-13
NFR-06	Performance	Search ≤ 3 s, other actions ≤ 5 s (50 users)	M	Product	FR-06
NFR-07	Reliability	No data lost or half saved after a sudden failure	M	Product	FR-09, FR-20
NFR-08	Compatibility	Same behavior on Windows 10 and 11	M	Organizational	FR-21, FR-29

2.5 Quality objectives

Table 2.4: Quality Objectives

Attribute	Objective	Target	Measurement	NFR
Usability	New customers book easily on their own	≥ 80% of users finish within 5 minutes	Usability test (7.7)	NFR-01
Security	No leak of passwords or personal data	100% of unauthorized access blocked	Authorization and database check	NFR-02, 03, 04
Maintainability	Rule changes need no code change	0 source files edited	Change test on the configuration	NFR-05
Performance	Stay fast with many users	≥ 95% of requests meet the time limit	Load test (50 users)	NFR-06

Attribute	Objective	Target	Measurement	NFR
Reliability	Failures cause no data loss	0 faulty or lost records after 10 crash tests	Crash test during saving	NFR-07
Compatibility	Works on Windows	100% of use cases pass on Windows 10 and 11	Test run on 2 operating systems	NFR-08

2.6 Assessment of whether the requirements are met

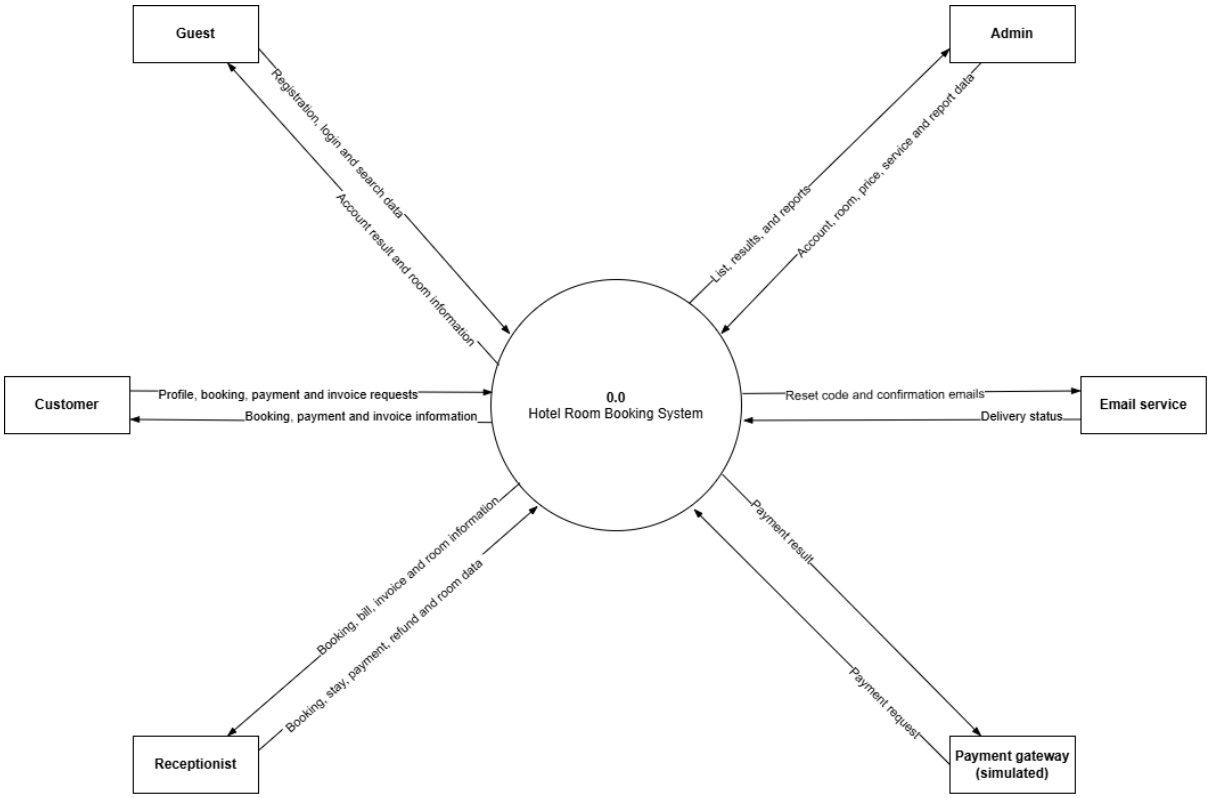
Status tracking at the design stage:

Table 2.5:Assessment of NFR Status at the Design Stage

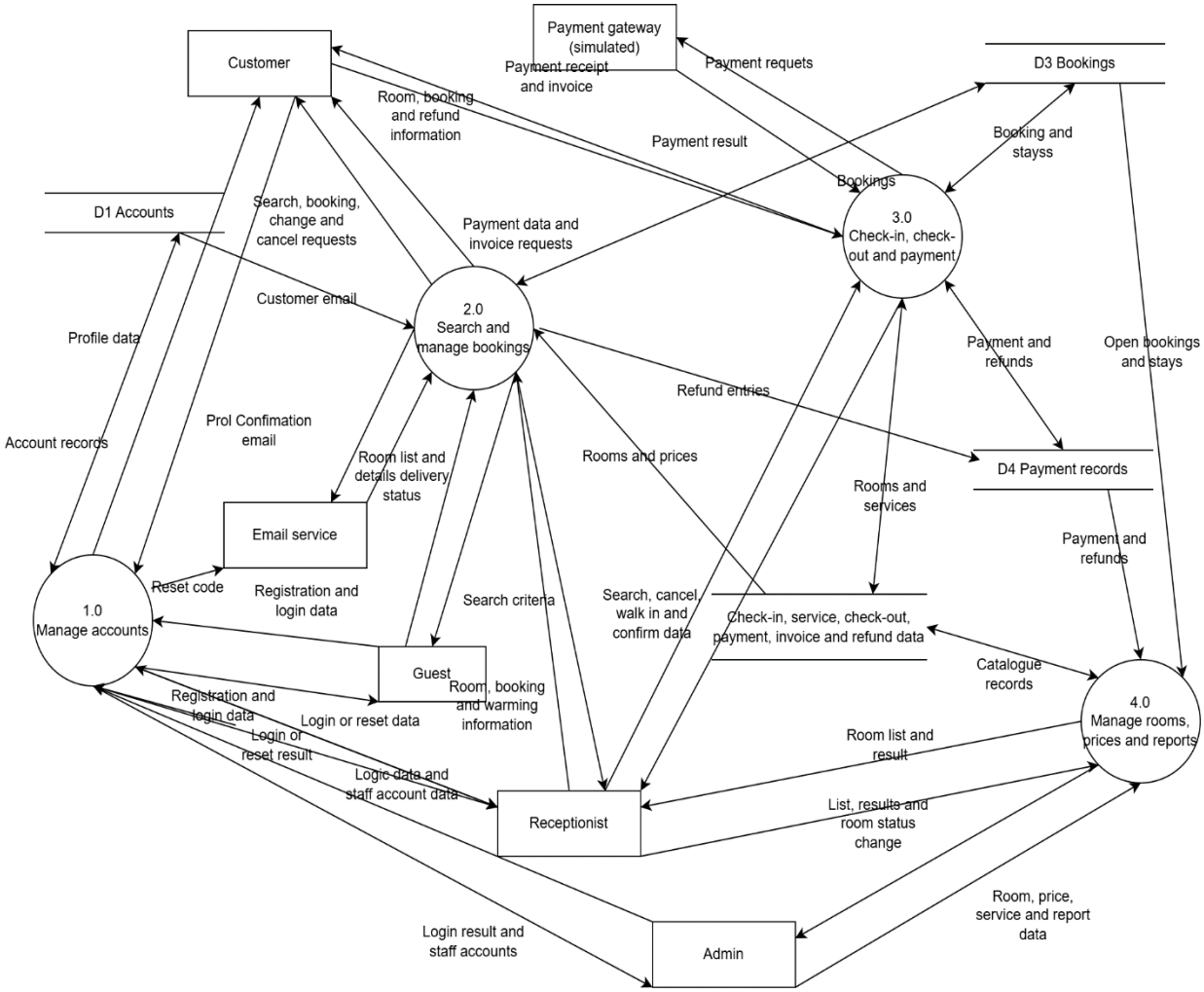
NFR	Supported by (Chapter 1)	Verified by	Status
NFR-01	Form pre-filled from the profile (FR-09), filters kept (FR-07)	Usability test (7.7)	Supported
NFR-02	Matrix 1.5, server-side check (CR-04)	Authorization test	Supported
NFR-03	Hashing rule BR-02, masking CR-08	Database and log inspection	Supported
NFR-04	Limited data fields (FR-01, 15, 16)	Field trace and test	Partly supported
NFR-05	Business rule list 1.4	Change test	Open (awaiting Ch. 5/6)
NFR-06	Time limit in FR-06	Load test	Open (awaiting Ch. 6)
NFR-07	Room hold (FR-10), email retry (FR-30)	Crash test	Partly supported
NFR-08	Platform assumption 1.1.3	Test run on 2 operating systems	Open (awaiting 5.6.4)

CHAPTER 3. DATA FLOW DIAGRAM (DFD)

3.1 DFD Level 0 (Context Diagram)

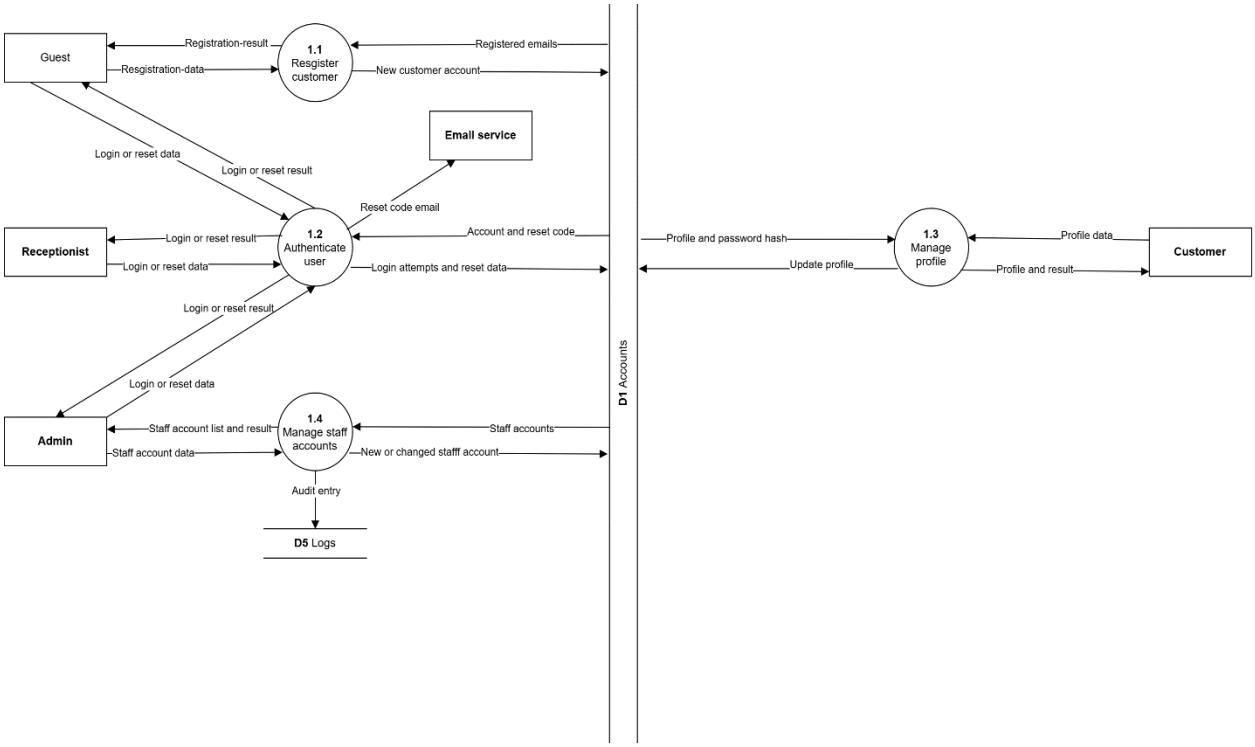


3.2 DFD Level 1

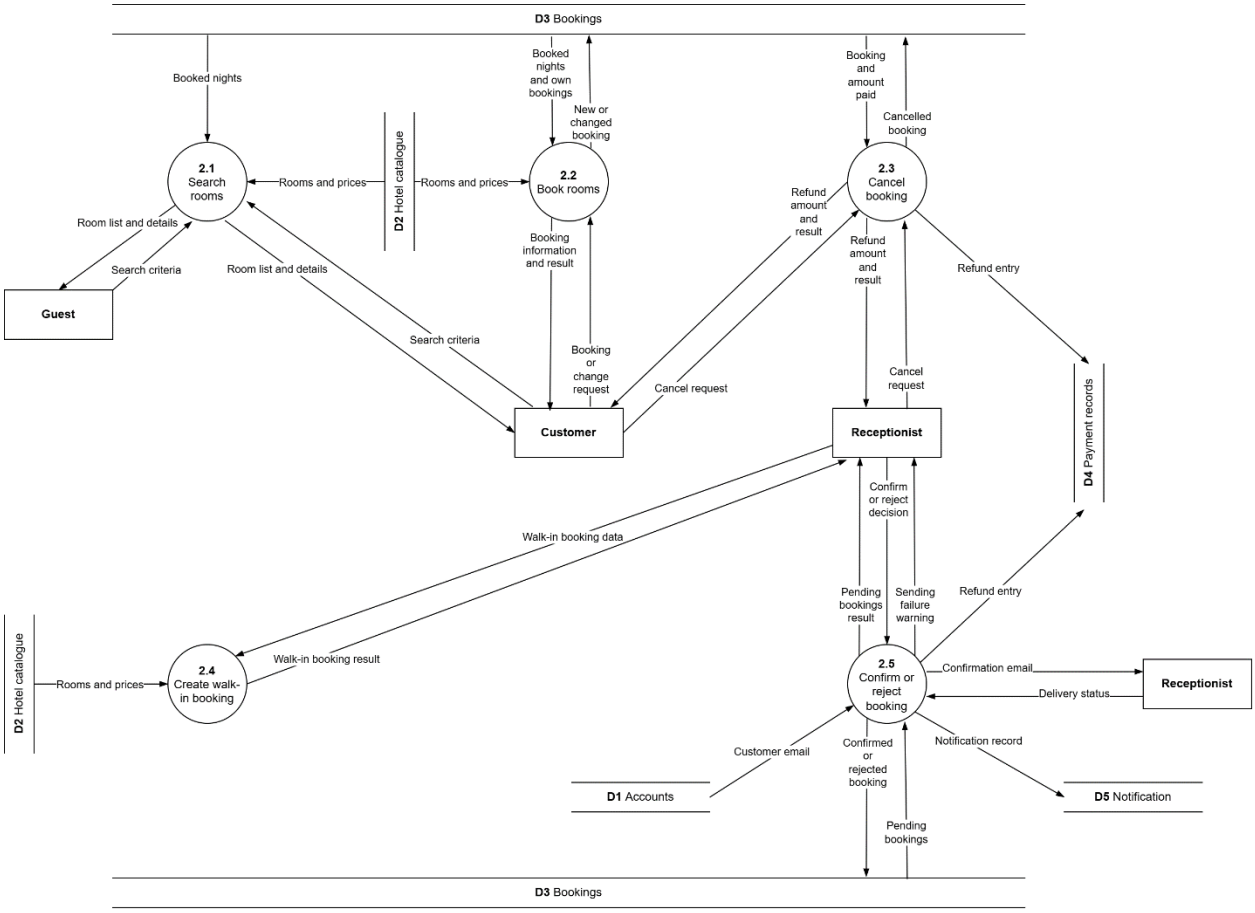


3.3 DFD Level 2

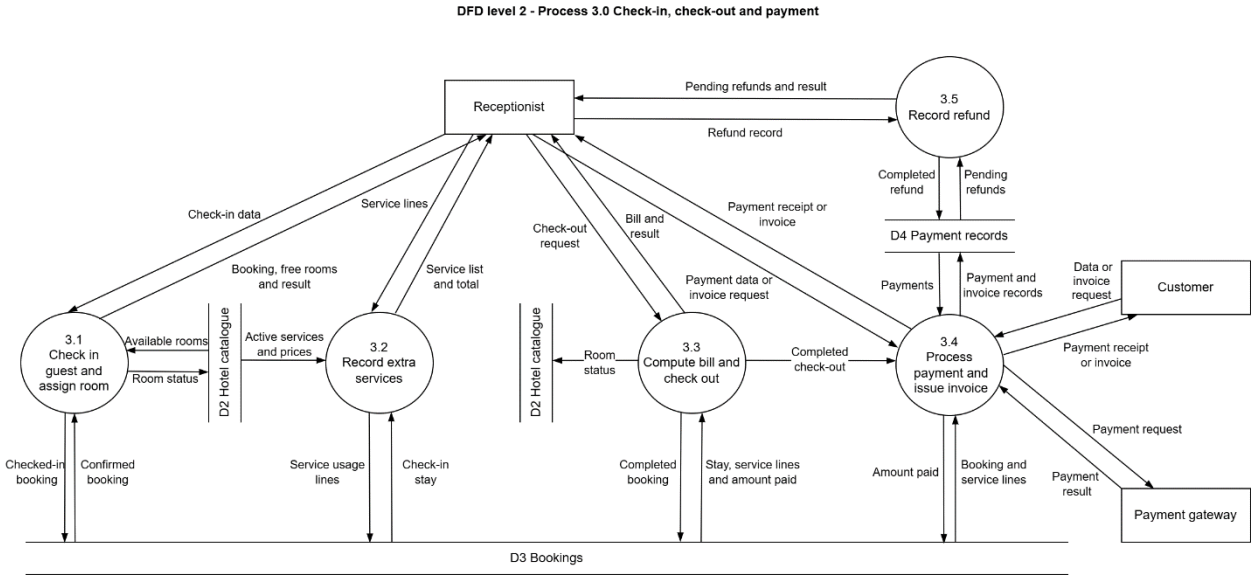
3.3.1 Process 1.0 - Account Management



3.3.2 Process 2.0 - Booking Management

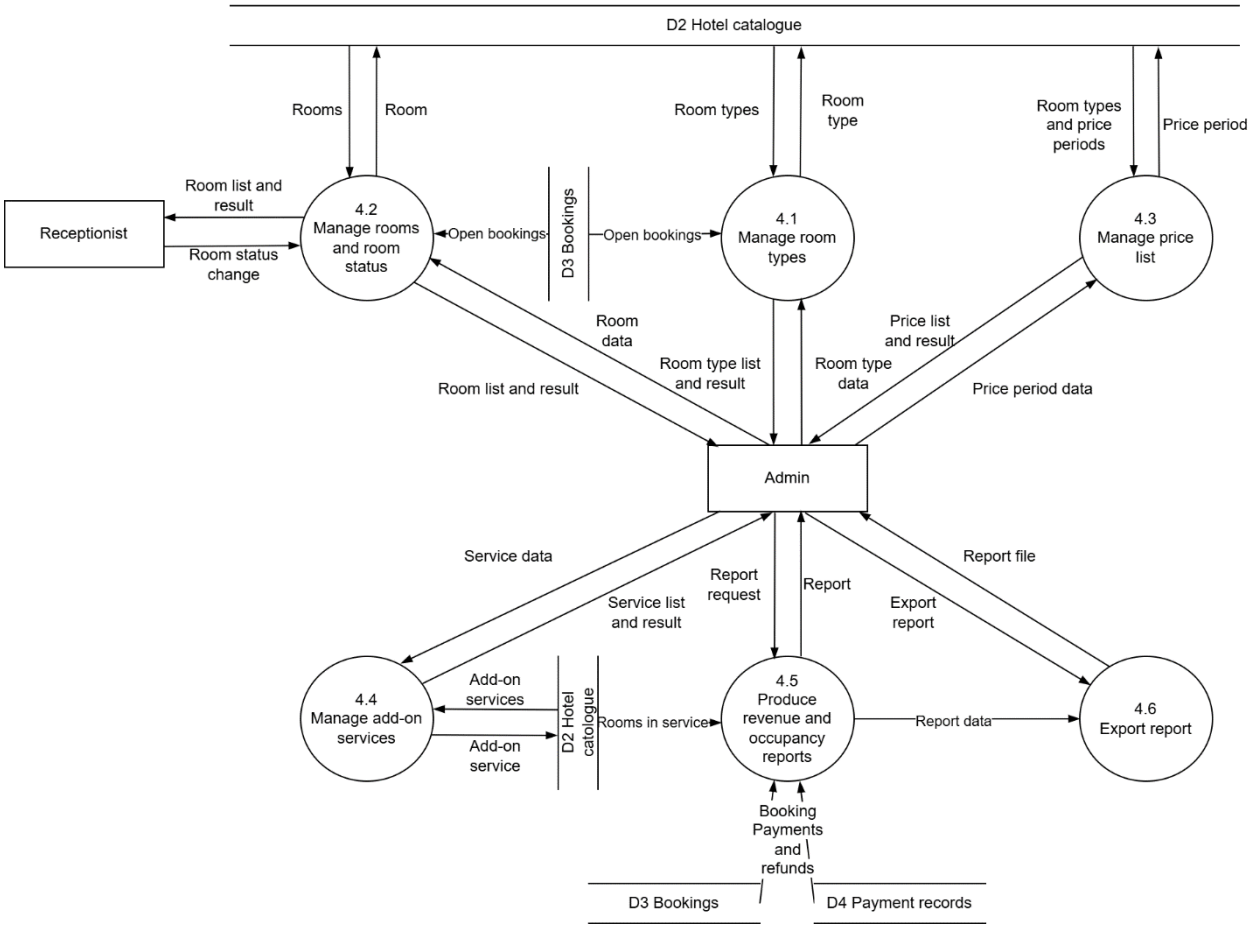


3.3.3 Process 3.0 - Stay & Front Desk Management



3.3.4 Process 4.0 - Catalog & Report Management

DFD level 2 - Process 4.0 Manage rooms, prices and reports



CHAPTER 4. USE CASE DIAGRAM AND SPECIFICATIONS

4.1 Actor Table and Overview Diagrams

Primary Actors: Guest, Customer (inherits Guest), Receptionist, Admin.

Secondary Actors: Email service, Payment gateway (simulated).

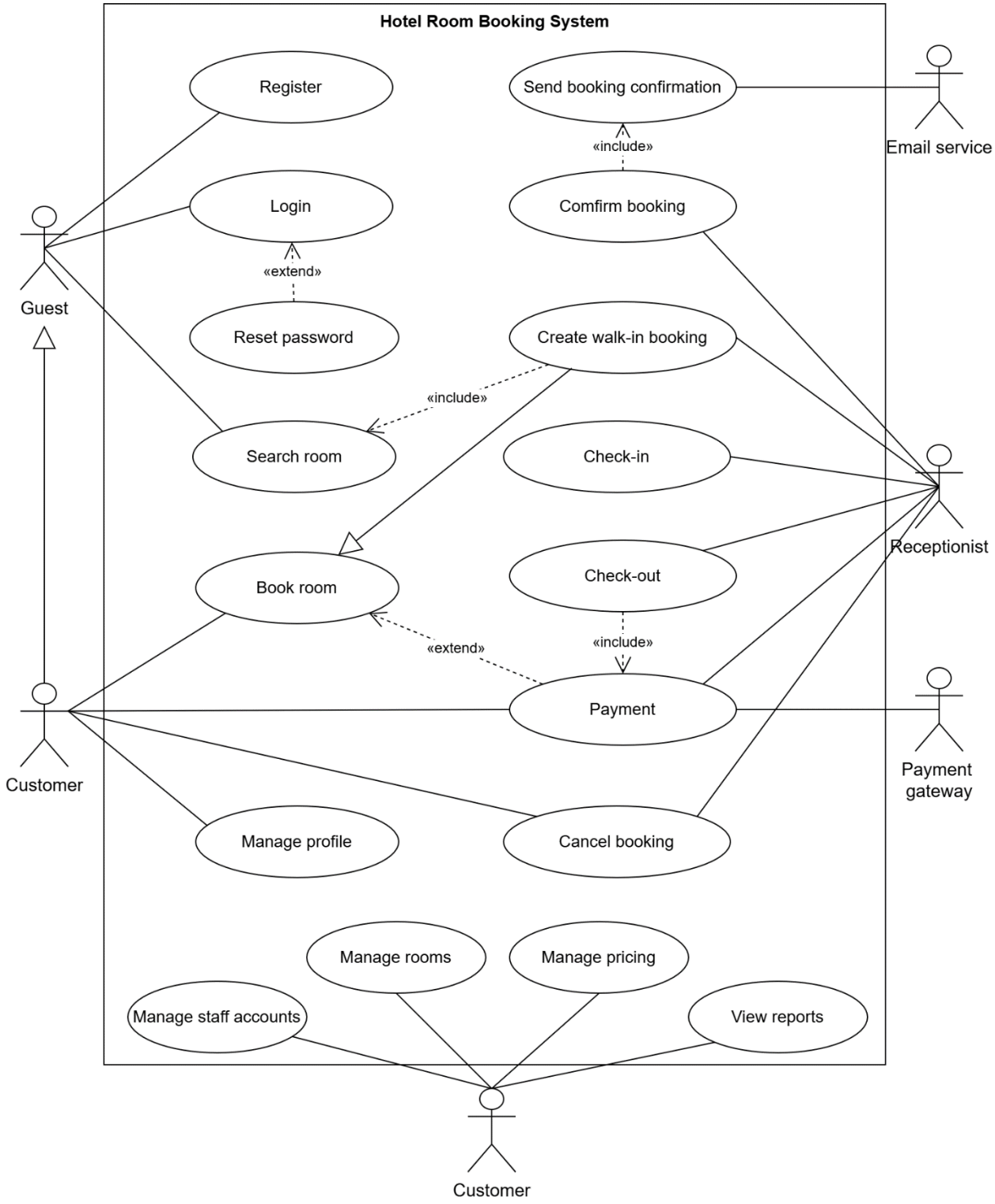


Figure 4.1: Overall Use Case Diagram



Figure 4.2: Use Cases of Guest and Customer

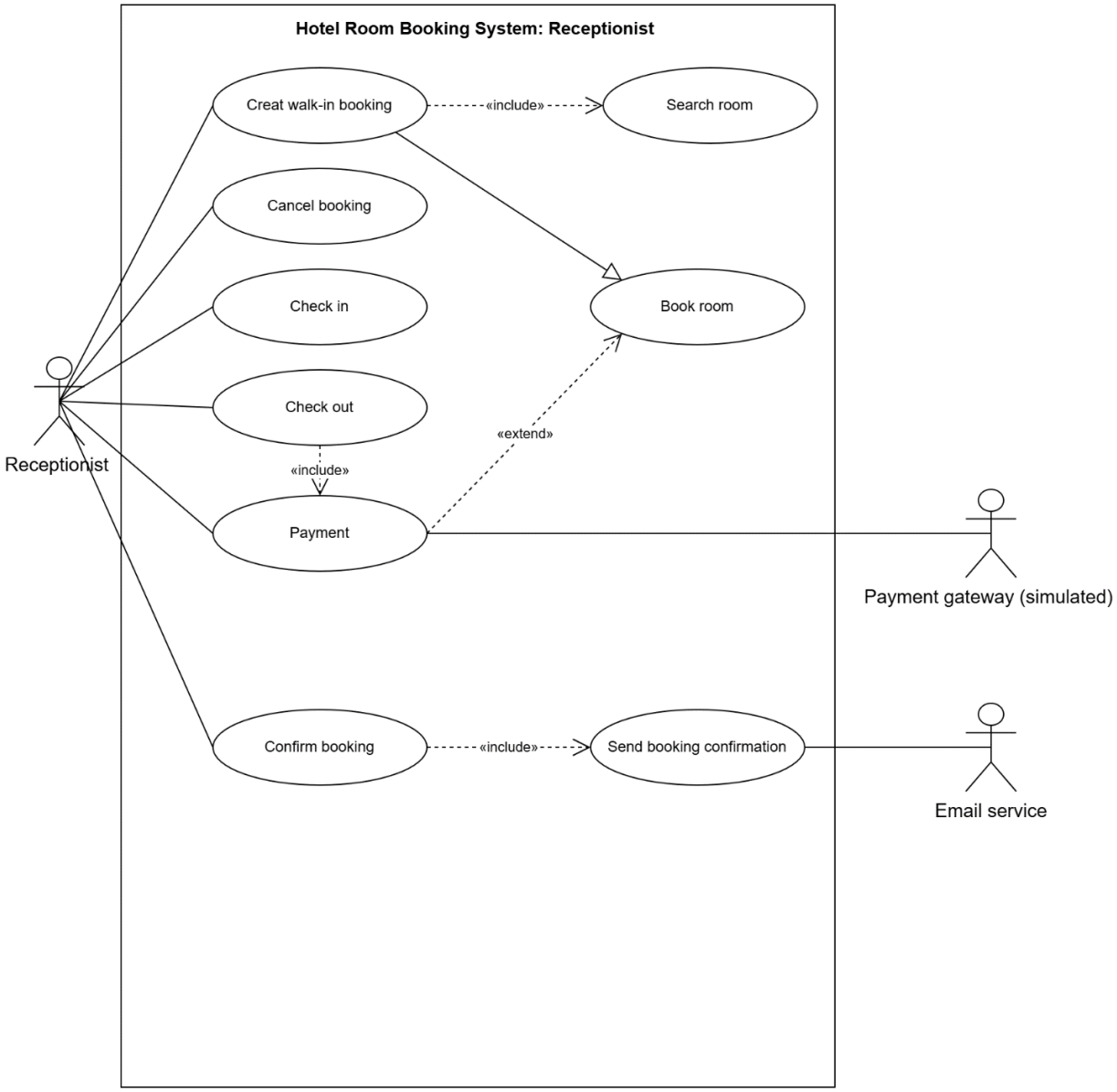


Figure 4.3: Use Cases of Receptionist

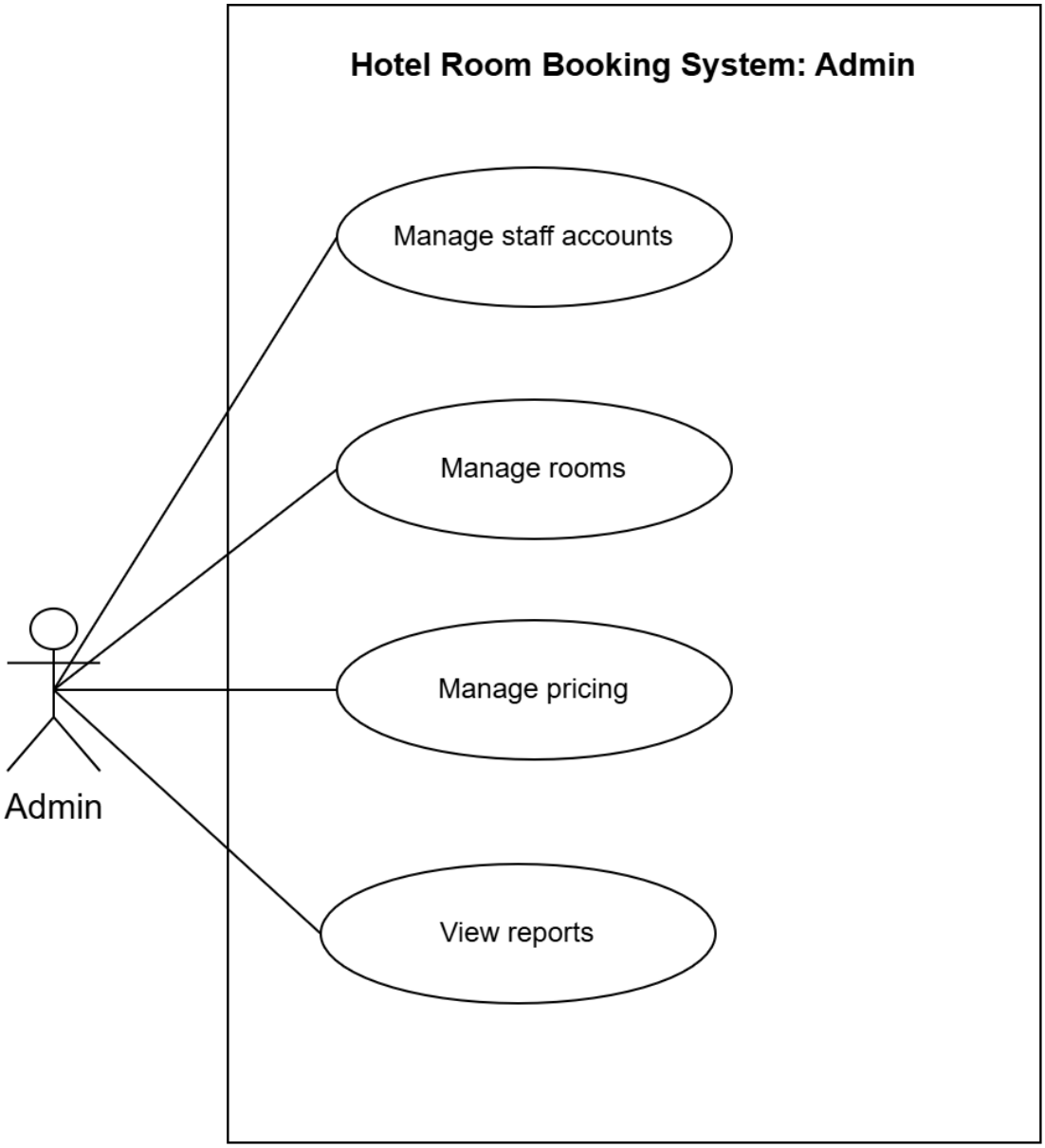


Figure 4.4: Use Cases of Admin

4.2 Use Case List and Traceability Matrix (17 Use Cases)

Tables 4.1: Use Case List and Traceability Matrix

ID	Use Case Name	Type	Primary Actor	Requirements	Pri	Screen
UC-01	Register	Main	Guest	FR-01	Must	SCR-02
UC-02	Login	Main	All Roles	FR-02	Must	SCR-01
UC-03	Search room	Main	Guest, Customer	FR-06, FR-07, FR-08	Must	SCR-05/06/07

ID	Use Case Name	Type	Primary Actor	Requirements	Pri	Screen
UC-04	Book room	Main	Customer	FR-09, FR-10, FR-11, FR-13	Must	SCR-08/11/12
UC-05	Cancel booking	Main	Customer, Receptionist	FR-12, FR-22	Must	SCR-12/14
UC-06	Confirm booking	Main	Receptionist	FR-14	Must	SCR-14
UC-07	Check-in	Main	Receptionist	FR-16, FR-17	Must	SCR-16
UC-08	Check-out	Main	Receptionist	FR-18, FR-19	Must	SCR-17/18
UC-09	Payment	Main	Customer, Receptionist	FR-20, FR-21	Must	SCR-09/18
UC-10	Manage rooms	Main	Admin	FR-23, FR-24, FR-26	Must	SCR-21/22/24
UC-11	Manage pricing	Main	Admin	FR-25	Must	SCR-23
UC-12	View reports	Main	Admin	FR-27, FR-28, FR-29	Should	SCR-25/26
UC-13	Manage profile	Support	Customer	FR-04	Should	SCR-04
UC-14	Reset password	Support	All Roles	FR-03	Should	SCR-03
UC-15	Send booking confirmation	Support	System (Email Service)	FR-30	Should	SCR-10
UC-16	Create walk-in booking	Support	Receptionist	FR-15	Should	SCR-15
UC-17	Manage staff accounts	Support	Admin	FR-05	Must	SCR-20

4.3 Detailed Use Case Specifications (All 17 Use Cases)

UC-01: Register

- Info: Guest | FR-01 | SCR-02 | Pre: Not logged in. Post: Customer account created.

- Basic flow: 1. Guest submits name, email, phone, password. 2. System validates fields, format, uniqueness (BR-01, BR-02). 3. System creates account with hashed password and displays MSG-01.

- Exceptions: 2a. Invalid input/duplicate email → Display MSG-02/04/05, retain typed inputs.

UC-02: Login

- Info: All Roles | FR-02 | SCR-01 | Pre: Valid account. Post: Active session, home screen loaded.

- Basic flow: 1. User enters email and password. 2. System verifies credentials and account status (BR-03). 3. System opens 30-min session (CR-01) and opens role-specific home screen.

- Exceptions: 2a. Incorrect credentials → Increment failed count, display MSG-06. 5th failure locks account (MSG-07).

UC-03: Search Room

- Info: Guest, Customer | FR-06, FR-07, FR-08 | SCR-05, SCR-06, SCR-07 | Pre: None. Post: Available rooms listed.

- Basic flow: 1. User enters dates, guests, optional room type. 2. System validates constraints (BR-06, BR-08). 3. System returns available room types within 3s (BR-07). 4. User filters/sorts results (FR-07). 5. User views room details (BR-12).

- Exceptions: 2a. Invalid dates/guests → MSG-14..17. 3a. No match → MSG-18. 5a. Guest clicks Book → Trigger UC-02 Login.

UC-04: Book Room

- Info: Customer | FR-09, FR-10, FR-11, FR-13 | SCR-08, SCR-11, SCR-12 | Pre: Logged in, room selected. Post: Booking Pending, room held 24h.

- Basic flow: 1. Customer clicks Book on room detail page. 2. System populates customer data and summary (30% deposit - BR-13). 3. Customer submits form. 4. System checks availability, holds room for 24h, generates code (HB-YYYYMMDD-NNNN). 5. System displays confirmation (SCR-10).

- Exceptions: 4a. Room taken concurrently → MSG-19. 5a. Deposit unpaid in 24h → Booking Expired, room released (BR-10, MSG-21).

UC-05: Cancel Booking

- Info: Customer, Receptionist | FR-12, FR-22 | SCR-12, SCR-14 | Pre: Booking Pending/Confirmed. Post: Booking Cancelled, refund calculated.

- Basic flow: 1. User selects booking and clicks Cancel. 2. System calculates refund based on policy (BR-12). 3. User confirms dialog (MSG-24). 4. System sets status to Cancelled, releases room, logs audit.

- Exceptions: 1a. Booking Checked-in/Completed → MSG-26. 4a. Receptionist records refund via FR-22 (MSG-44).

UC-06: Confirm Booking

- Info: Receptionist | FR-14 | SCR-14 | Pre: Booking Pending. Post: Booking Confirmed or Rejected.

- Basic flow: 1. Receptionist reviews Pending list on SCR-14. 2. Receptionist verifies deposit payment. 3. Receptionist clicks Confirm → Status set to Confirmed, triggers UC-15. (Or clicks Reject with reason → Status Rejected, room released, refund queued).

- Exceptions: 2a. Booking expired → MSG-21. 3a. Reject without reason → MSG-29.

UC-07: Check-in

- Info: Receptionist | FR-16, FR-17 | SCR-16 | Pre: Booking Confirmed, arrival date reached. Post: Booking Checked-in, Room Occupied.

- Basic flow: 1. Receptionist searches Confirmed booking. 2. Receptionist enters guest ID document number. 3. System lists available rooms of booked type. 4. Receptionist assigns room → Booking set to Checked-in, room set to Occupied.

- Exceptions: 1a. Date not reached / Not confirmed → MSG-31. 3a. No room Available → MSG-32. 4a. Room swap during stay → Set old room Cleaning, new room Occupied, log audit (MSG-34).

UC-08: Check-out

- Info: Receptionist | FR-18, FR-19 | SCR-17, SCR-18 | Pre: Guest Checked-in. Post: Booking Completed, Room Cleaning, Invoice generated.

- Basic flow: 1. Receptionist records extra services during stay (SCR-17). 2. At departure, Receptionist initiates check-out on SCR-18. 3. System computes bill: Room + Services + Late fee (>12:00 adds 50% night) + 10% VAT - Deposit. 4. Receptionist collects balance via UC-09. 5. System completes check-out, sets room to Cleaning, generates invoice.

- Exceptions: 4a. Unpaid balance → Block check-out, display MSG-38.

UC-09: Payment

- Info: Customer, Receptionist | FR-20, FR-21 | SCR-09, SCR-18 | Pre: Balance > 0. Post: Payment recorded, invoice issued if check-out.

- Basic flow: 1. User selects payment method (Cash, Card, E-Wallet). 2. System validates amount ≥ minimum deposit. 3. Simulated gateway responds success. 4. System records transaction, updates amount paid, issues invoice (INV-YYYYMMDD-NNNN) if full check-out.

- Exceptions: 3a. Gateway failure → MSG-41. 4a. PDF creation fail → MSG-58 (can reprint).

UC-10: Manage Rooms

- Info: Admin | FR-23, FR-24, FR-26 | SCR-21, SCR-22, SCR-24 | Pre: Admin logged in. Post: Inventory updated.

- Basic flow: 1. Admin selects Room Types, Rooms, or Add-on Services. 2. Admin adds, edits, or deactivates items. 3. System validates unique names/numbers and records changes to audit log.

- Exceptions: 2a. Delete item with active/future bookings → Blocked, display MSG-49. 2b. Set Occupied room to Maintenance → Blocked, display MSG-51.

UC-11: Manage Pricing

- Info: Admin | FR-25 | SCR-23 | Pre: Admin logged in, room types exist. Post: Price period saved.

- Basic flow: 1. Admin enters room type, start date, end date, and price per night. 2. System verifies non-overlapping date periods for same room type (BR-31). 3. System saves period and logs audit.

- Exceptions: 2a. Overlapping date period → Display MSG-50. 2b. Invalid price ≤ 0 → Display MSG-59.

UC-12: View Reports

- Info: Admin | FR-27, FR-28, FR-29 | SCR-25, SCR-26 | Pre: Admin logged in. Post: Reports generated/exported.

- Basic flow: 1. Admin selects report type (Revenue/Occupancy), date range, and grouping (day/month). 2. System computes aggregates (BR-33, BR-34) and displays table + chart. 3. Admin exports to PDF/Excel (FR-29).

- Exceptions: 1a. End date before start date → MSG-52. 2a. No data → MSG-53.

UC-13: Manage Profile

- Info: Customer | FR-04 | SCR-04 | Pre: Customer logged in. Post: Profile updated.

- Basic flow: 1. Customer views profile. 2. Customer updates name, phone, or password (requires current password). 3. System saves changes and displays MSG-11.

- Exceptions: 2a. Incorrect current password → MSG-12.

UC-14: Reset Password

- Info: All Roles | FR-03 | SCR-03 | Pre: User forgot password. Post: Password updated via code.

- Basic flow: 1. User requests reset code via email. 2. System sends 30-min single-use code (BR-04). 3. User enters code and new password. 4. System updates password hash and displays MSG-10.

- Exceptions: 3a. Invalid/expired code → MSG-09.

UC-15: Send Booking Confirmation

- Info: System (Email Service) | FR-30 | SCR-10 | Pre: Booking Confirmed. Post: Email dispatched.

- Basic flow: 1. System automatically generates confirmation details upon booking confirmation. 2. System sends email via Email Service. 3. If failed, system retries up to 3 times (BR-35).

- Exceptions: 3a. Email failure → Display MSG-55 to Receptionist, booking stays Confirmed.

UC-16: Create Walk-in Booking

- Info: Receptionist | FR-15 | SCR-15 | Pre: Receptionist logged in. Post: Booking Confirmed created without account.

- Basic flow: 1. Receptionist searches room availability (UC-03). 2. Receptionist enters guest name and phone number. 3. System creates booking with status Confirmed (BR-15). 4. Receptionist proceeds directly to Check-in (UC-07).

- Exceptions: 1a. No room available → MSG-19.

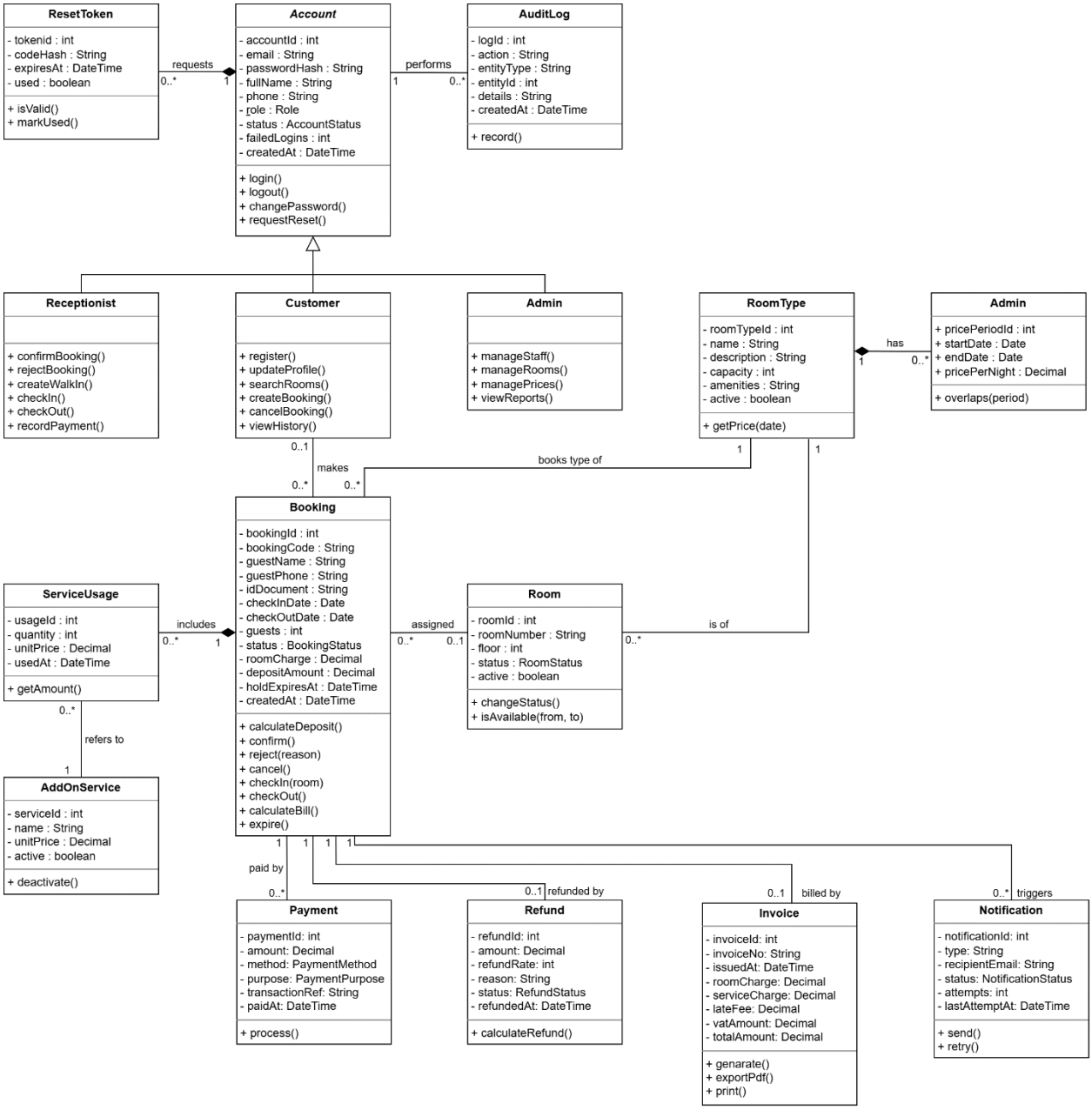
UC-17: Manage Staff Accounts

- Info: Admin | FR-05 | SCR-20 | Pre: Admin logged in. Post: Staff account created/updated.

- Basic flow: 1. Admin enters staff details, role (Receptionist/Admin), and status. 2. System saves staff account and logs audit. 3. Admin can lock/unlock or reset staff passwords.

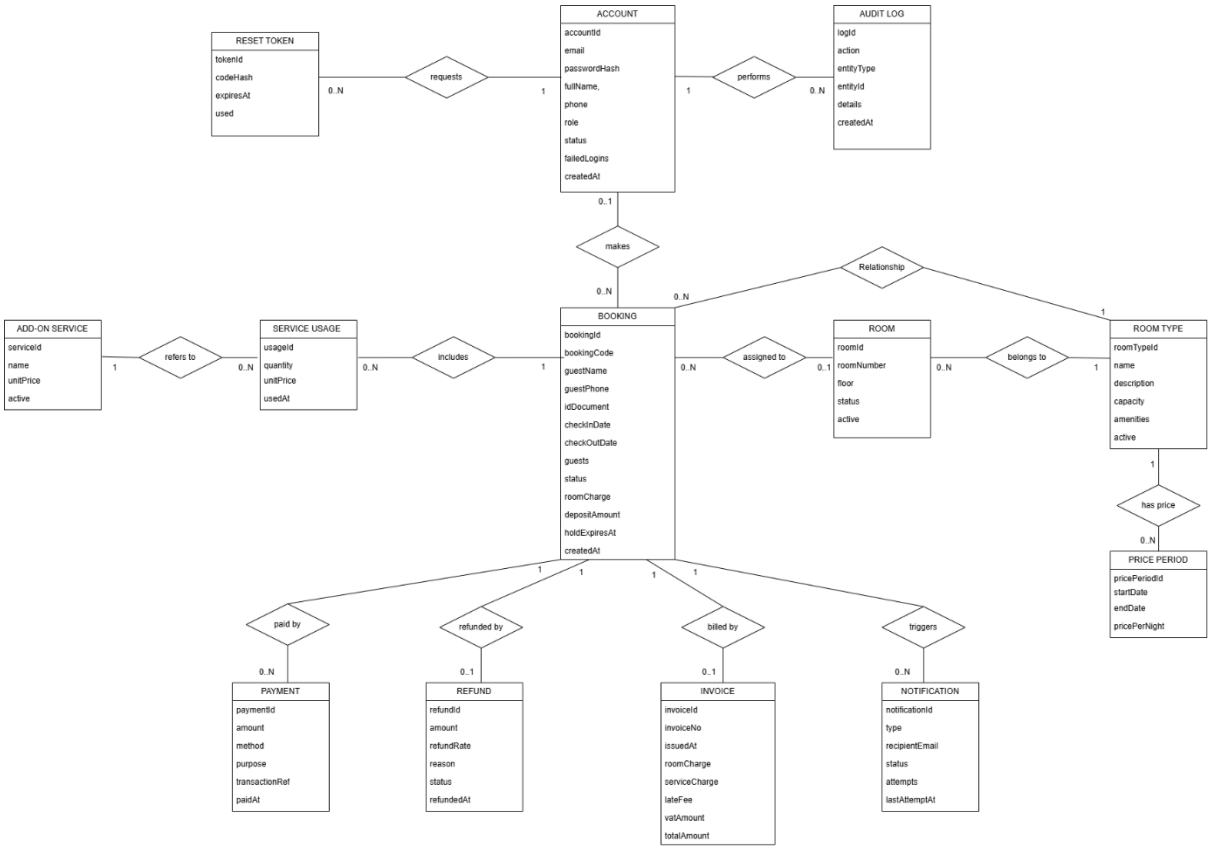
- Exceptions: 2a. Attempt to lock last active Admin → Blocked, display MSG-60.

CHAPTER 5. CLASS DIAGRAM

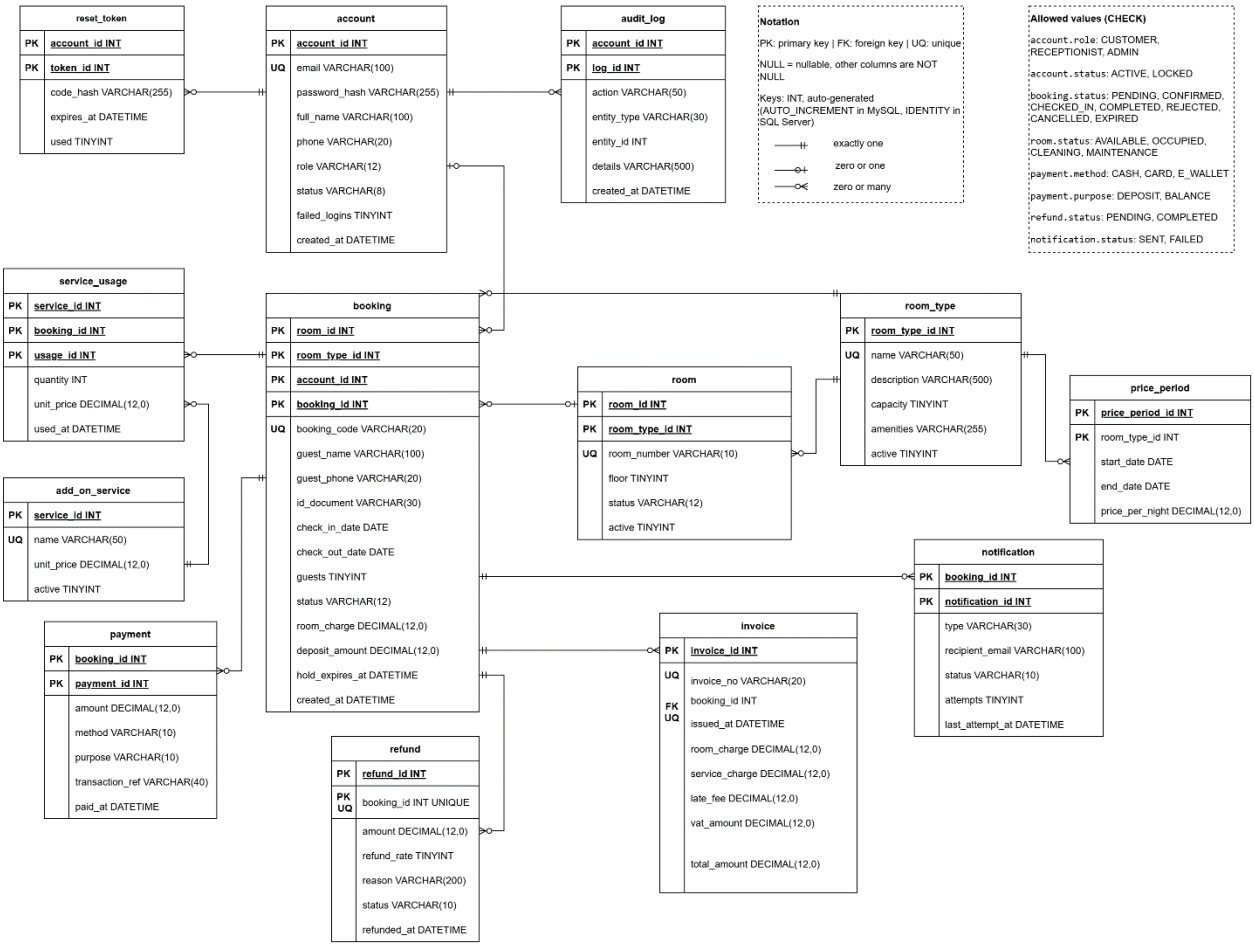


CHAPTER 6. DATA MODEL

6.1 Conceptual Data Model (ERD)



6.2 Logical Database Schema



CHAPTER 7. INTERFACE DESIGN DESCRIPTION

7.1 Guest and Customer Interfaces

7.1.1 SCR-01: Login Screen

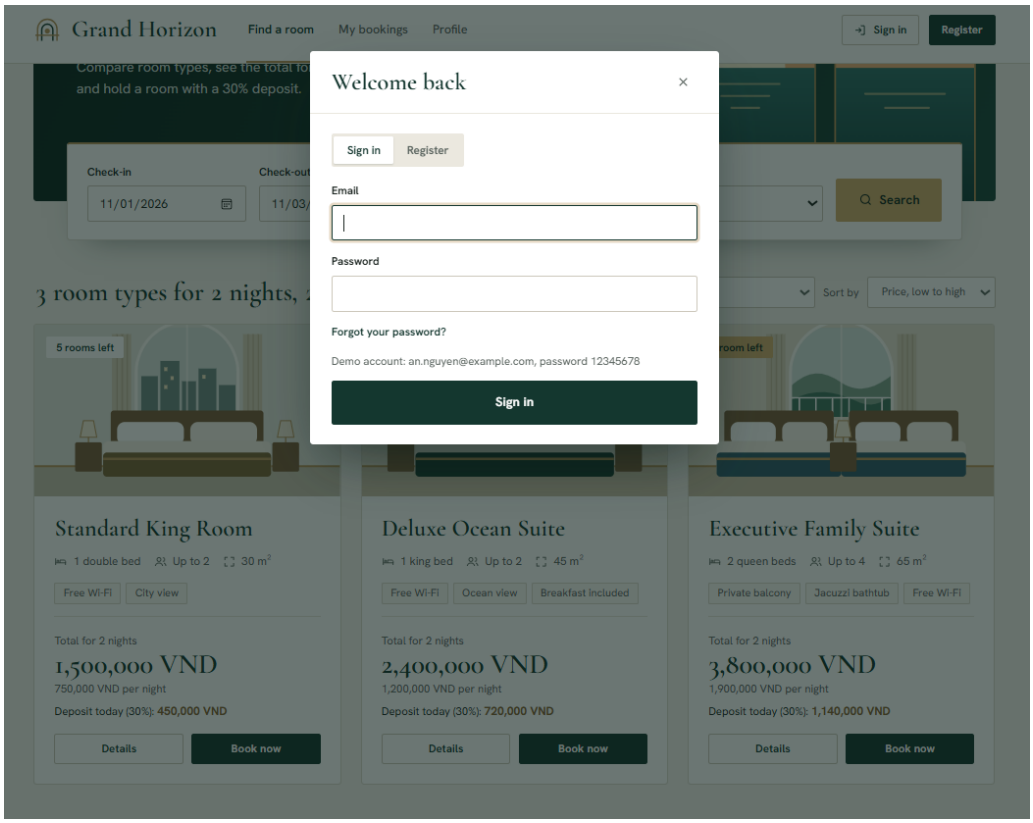


Figure 7.1: SCR-01 - Login Screen (Sign in)

Pop-up with Email and Password fields, a “Forgot your password?” link and a Sign in button; the Sign in, Register tabs let a visitor switch to account creation.

It opens from the Sign in button in the header and is shared by all roles; a wrong email or password shows the message “The email or password is not correct.”

7.1.2 SCR-02: Register Screen

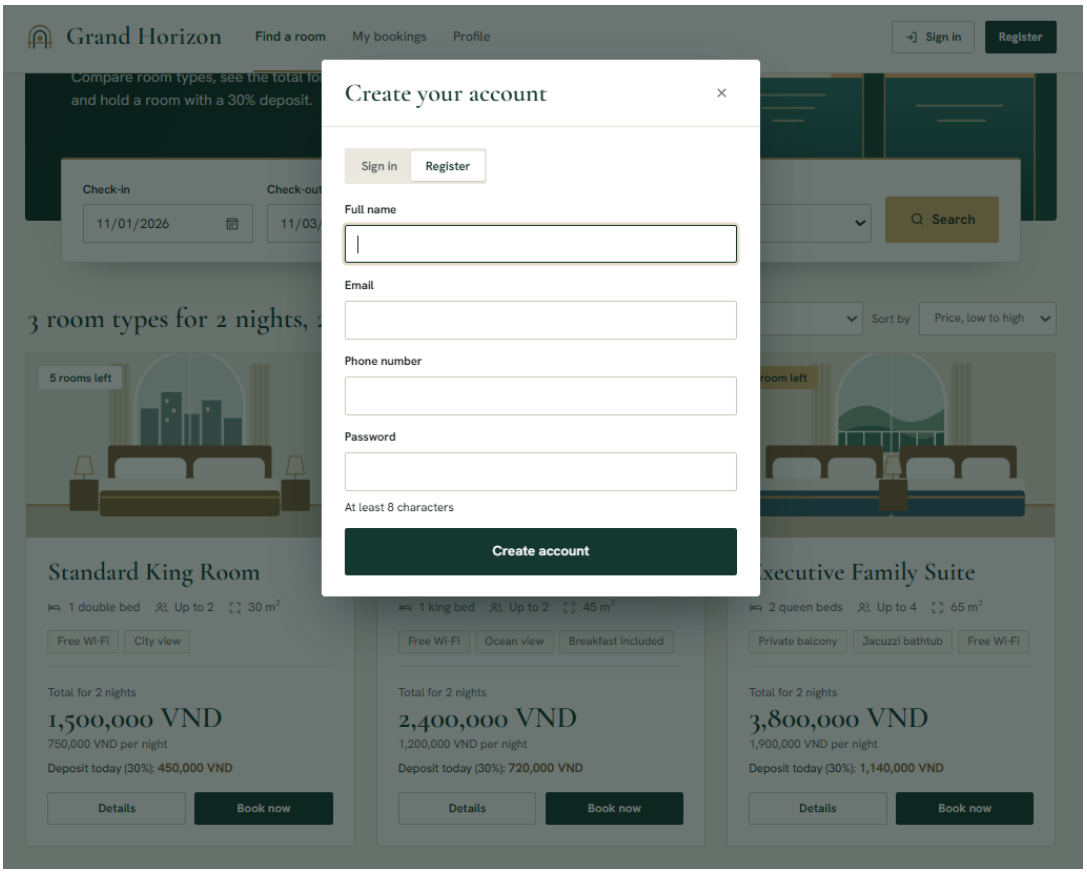


Figure 7.2: SCR-02 - Register Screen

Collects Full name, Email, Phone number and Password in one short form to create a Customer account.

Every field is validated before saving: a valid and unused email, a phone number of at least 9 digits and a password of at least 8 characters.

7.1.3 SCR-03: Reset Password Screen

Figure 7.3: SCR-03 - Reset Password Screen

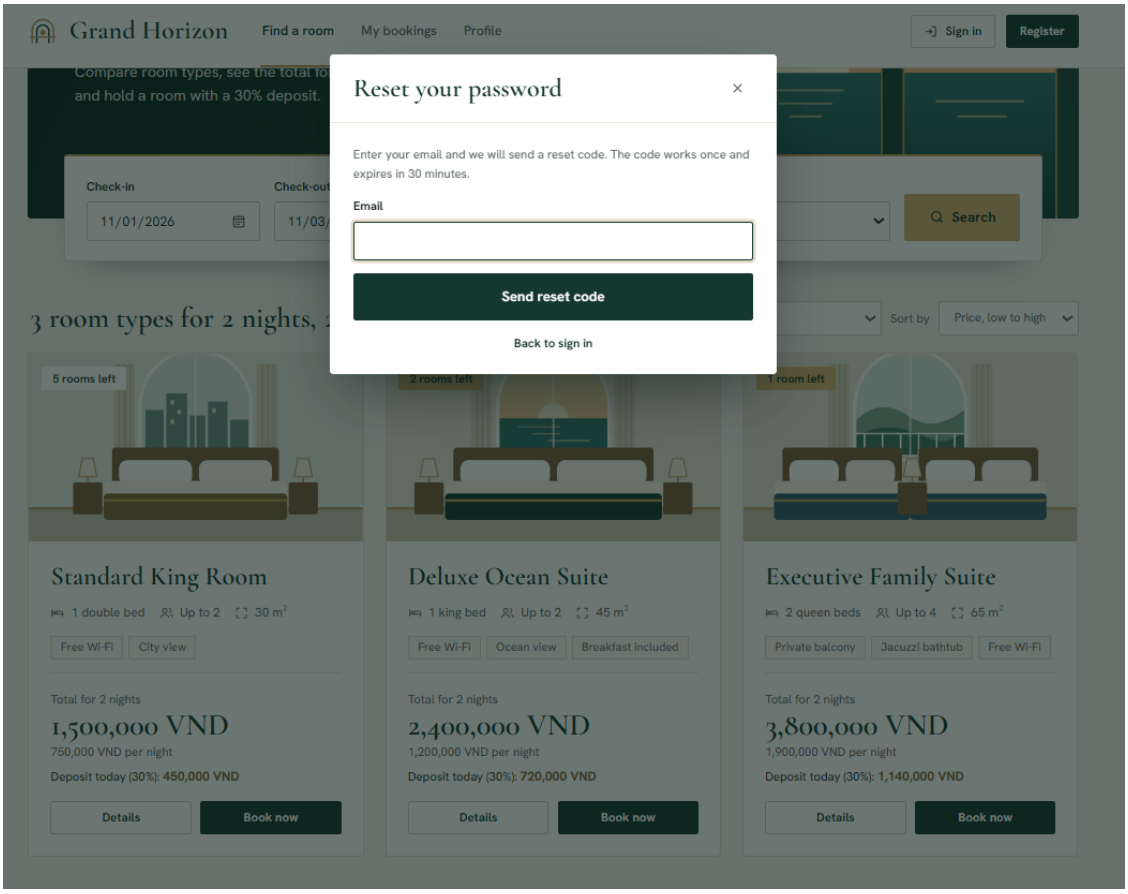


Figure 7.3: SCR-03 - Reset Password Screen

The user enters the account email and requests a reset code with the Send reset code button; Back to sign in returns to the login form.

The screen states that the code works once and expires after 30 minutes, so an old code cannot be reused.

7.1.4 SCR-04: Customer Profile Screen

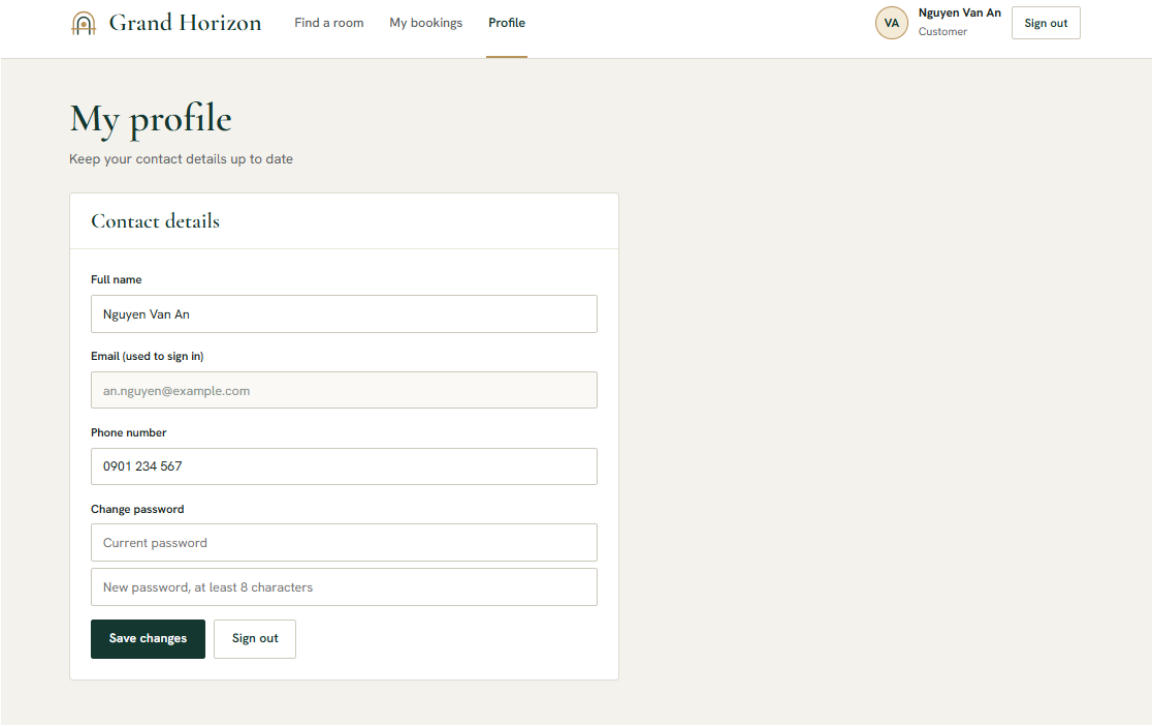


Figure 7.4: SCR-04 - Customer Profile Screen

Shows the Contact details of the signed-in customer: Full name and Phone number can be edited, while the sign-in Email is read-only.

The password is changed by entering the current password and a new one of at least 8 characters; Save changes updates the profile and Sign out ends the session.

7.1.5 SCR-05: Customer Homepage & Room Search

Figure 7.5: SCR-05 - Customer Homepage and Room Search

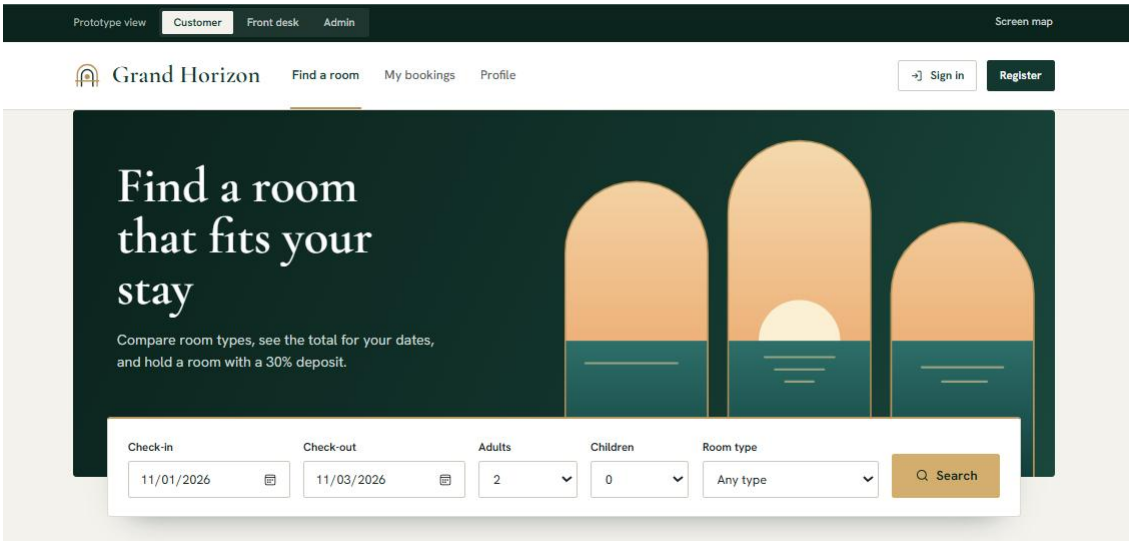


Figure 7.5: SCR-05 - Customer Homepage and Room Search

Landing page of the system with a search bar for Check-in, Check-out, Adults, Children and Room type; the Search button lists the matching room types.

It is open to guests who have not signed in, default dates are pre-filled, and the dates are validated (no past check-in, check-out after check-in, at most 30 nights).

7.1.6 SCR-06: Search Results & Filters

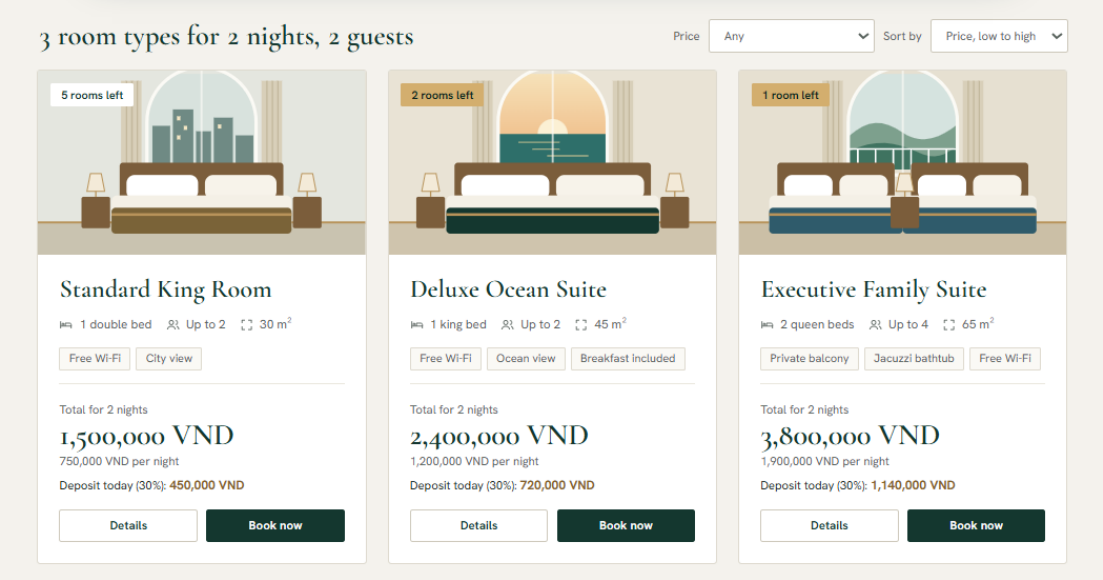


Figure 7.6: SCR-06 - Search Results and Filters

The result header summarises the number of room types, nights and guests, and the Price filter and Sort by list let the customer narrow and order the results.

Each room card shows a photo, the number of rooms left, bed, capacity, size, amenities, the total for the stay, the price per night and the 30% deposit, with Details and Book now buttons.

7.1.7 SCR-07: Room Detail Screen

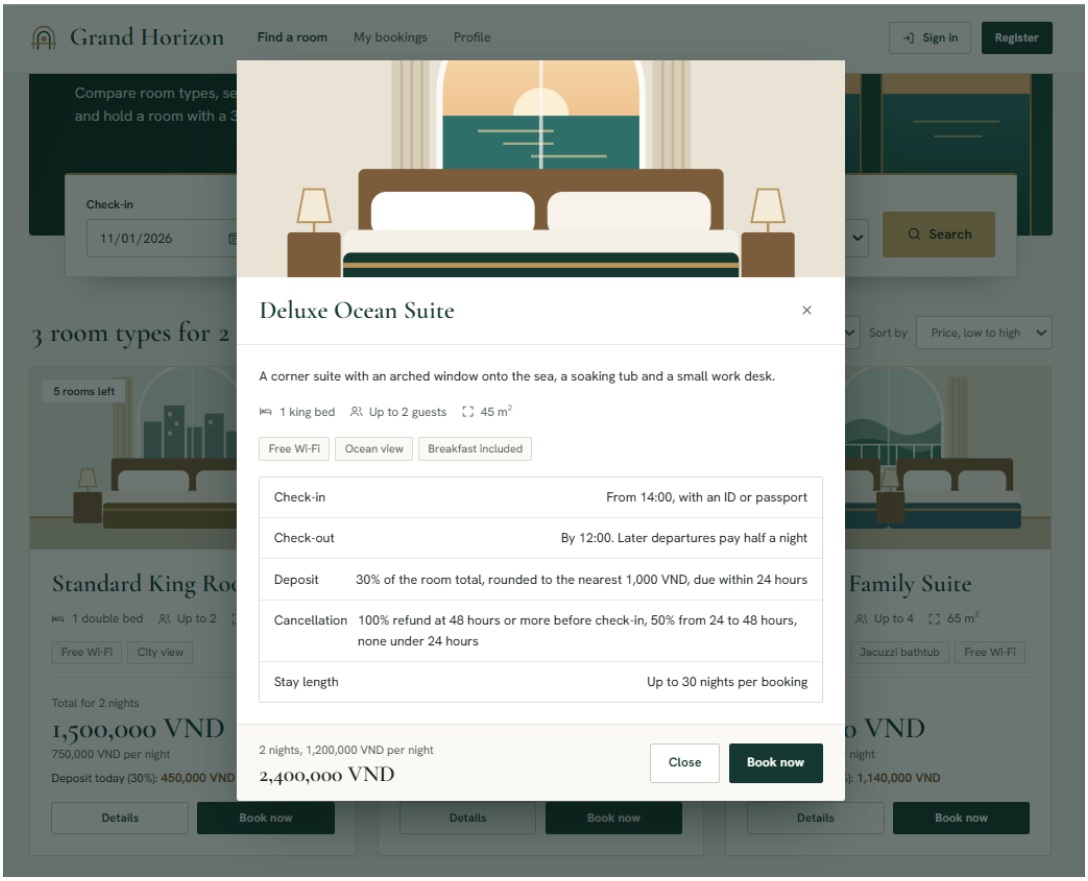


Figure 7.7: SCR-07 - Room Detail Screen

Pop-up with the full description of the room type: bed type, capacity, size, amenities and the total price for the selected stay, with a Book now button.

It also lists the policies: check-in from 14:00 with an ID or passport, check-out by 12:00, a 30% deposit due within 24 hours, the cancellation refund tiers (100%, 50% or none) and the 30-night limit.

7.1.8 SCR-08: Booking Form

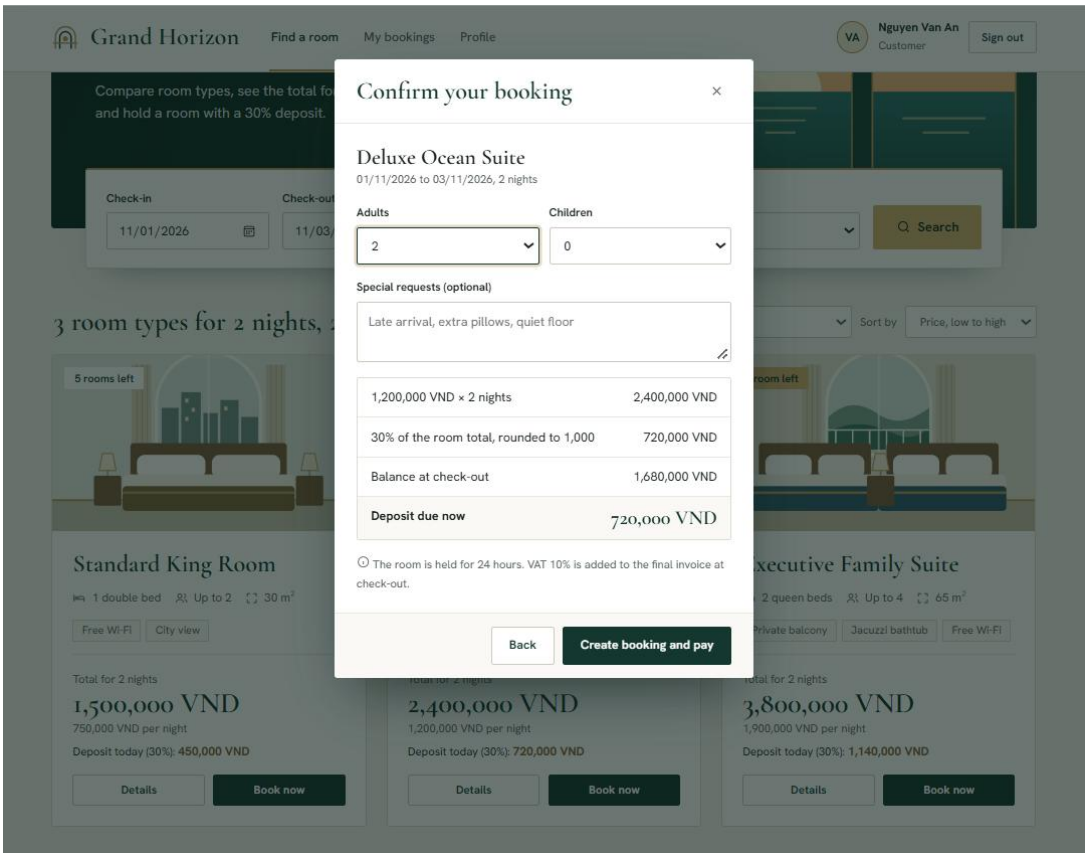


Figure 7.8: SCR-08 - Booking Form

Confirmation form for the chosen room type and dates, where the customer sets the number of Adults and Children (checked against the room capacity) and may add special requests.

It shows the price breakdown: room total, the 30% deposit rounded to the nearest 1,000 VND and the balance at check-out, with the notes that the room is held for 24 hours and VAT 10% is added at check-out.

7.1.9 SCR-09: Payment (Pay Deposit)

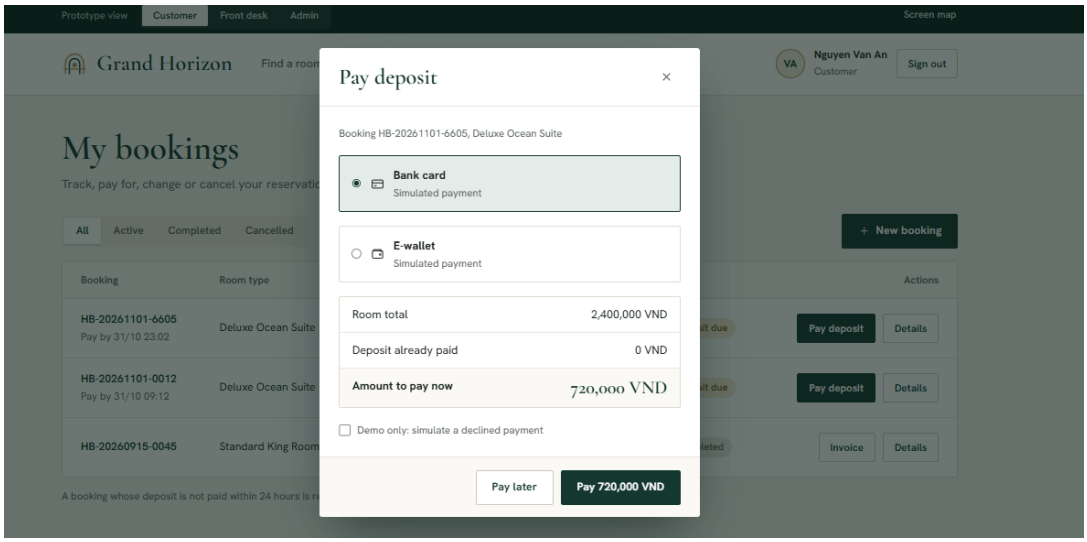


Figure 7.9: SCR-09 - Payment (Pay Deposit)

Deposit payment pop-up showing the booking code, the room total, the deposit already paid and the amount to pay now, the customer chooses Bank card or E-wallet (simulated payment).

Pay confirms the deposit, while Pay later leaves the booking unpaid, an unpaid booking is released automatically if the deposit is not paid within 24 hours.

7.1.10 SCR-10: Booking Confirmation Screen

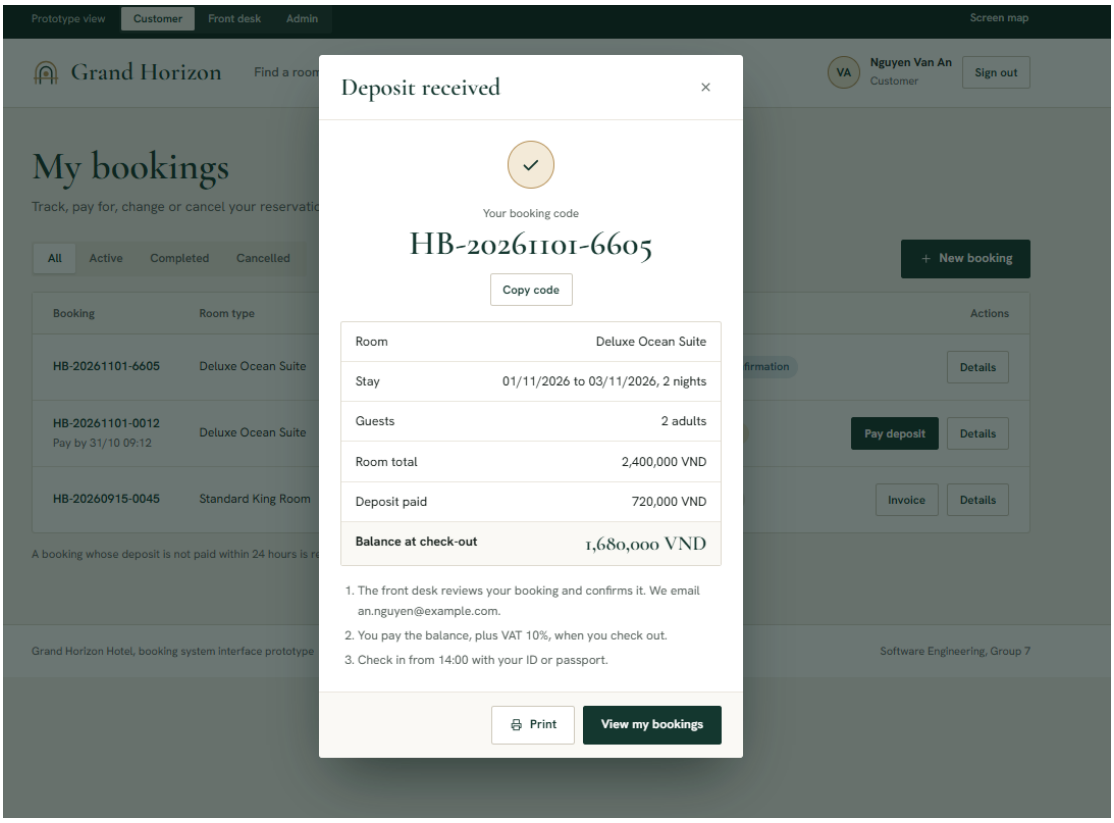


Figure 7.10: SCR-10 - Booking Confirmation Screen

Confirms that the deposit was received and shows the booking code with a Copy code button, together with the room, stay, guests, room total, deposit paid and balance at check-out.

It lists the next steps: the front desk reviews and confirms the booking, the balance plus 10% VAT is paid at check-out, and check-in is from 14:00 with an ID or passport; Print and View my bookings are available.

7.1.11 SCR-11: Booking History Screen

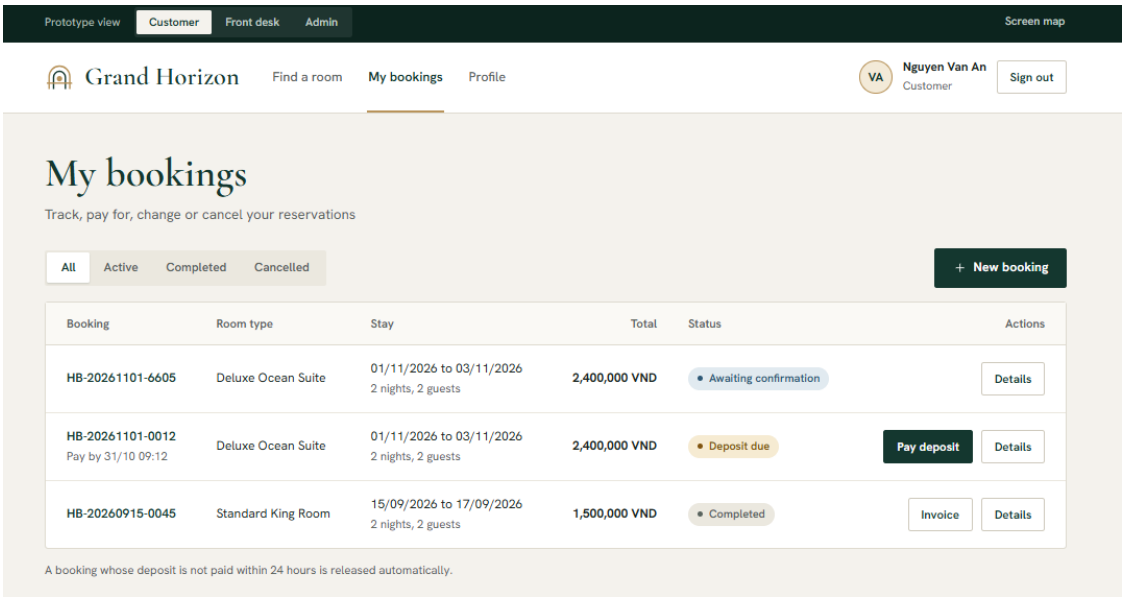


Figure 7.11: SCR-11 - Booking History Screen

Table of the customer's bookings with code, room type, stay, total, status and actions; the tabs All, Active, Completed and Cancelled filter the list.

The available action depends on the status: Pay deposit for unpaid bookings, Details for every booking and Invoice for completed stays.

7.1.12 SCR-12: Booking Details, Change & Cancel

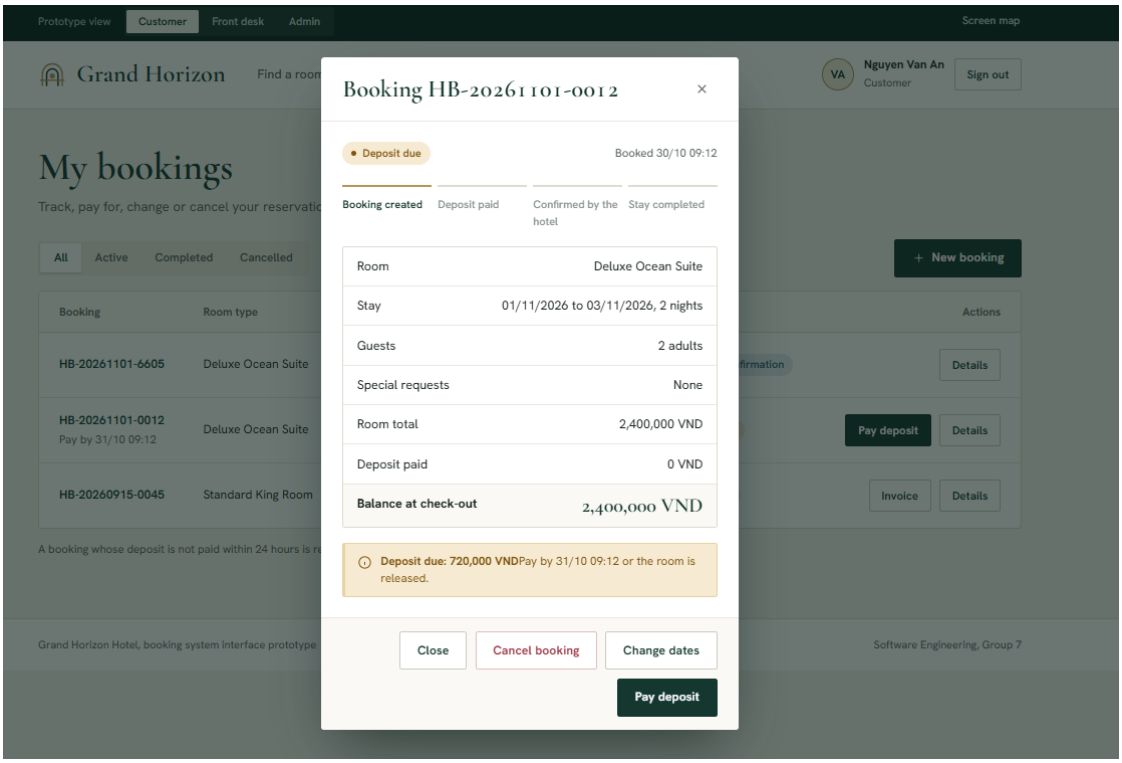


Figure 7.12: SCR-12 - Booking Details, Change and Cancel

Shows one booking with its status, a progress timeline (Booking created, Deposit paid, Confirmed by the hotel, Stay completed) and the room, stay, guests, totals and balance at check-out.

From here the customer can Pay deposit, Cancel booking or Change dates; a date change is accepted only if the new check-in is at least 48 hours away and rooms of that type are still available.

7.2 Receptionist Interfaces

7.2.1 SCR-13: Front Desk Dashboard

Figure 7.13: SCR-13 – Front Desk Dashboard

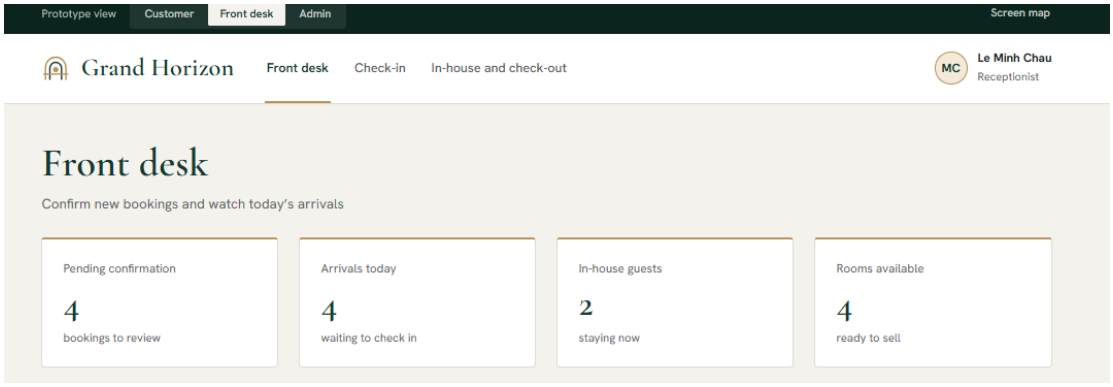


Figure 7.13: SCR-13 - Front Desk Dashboard

Landing page of the receptionist with four counters: Pending confirmation, Arrivals today, In-house guests and Rooms available.

The New walk-in booking button gives quick access to booking a room for a guest who arrives without a reservation.

7.2.2 SCR-14: Pending Bookings Screen

Figure 7.14: SCR-14 - Pending Bookings Screen

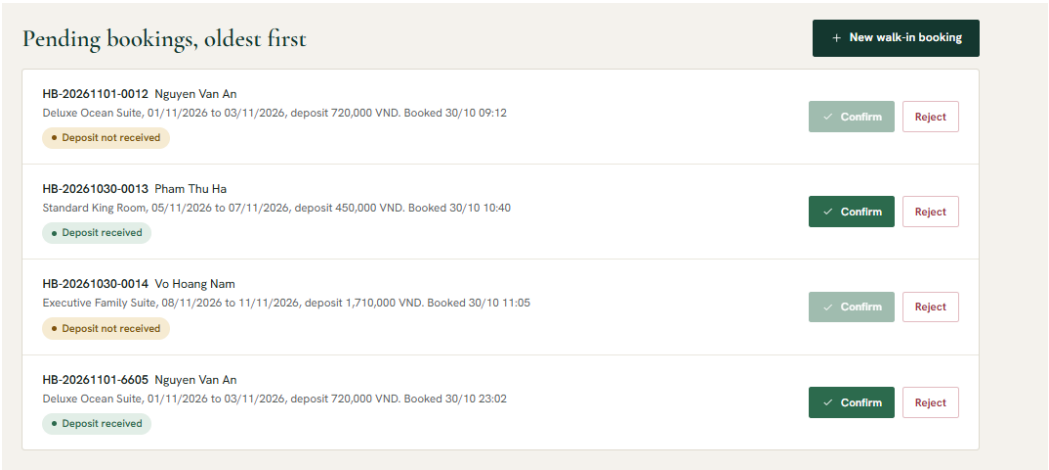


Figure 7.14: SCR-14 - Pending Bookings Screen

Queue of bookings waiting for review, oldest first, showing the booking code, guest, room type, dates, deposit amount and the time the booking was made.

A badge shows whether the deposit has been received; Confirm is enabled only after the deposit is received, and Reject declines the booking.

7.2.3 SCR-15: Walk-in Booking Screen

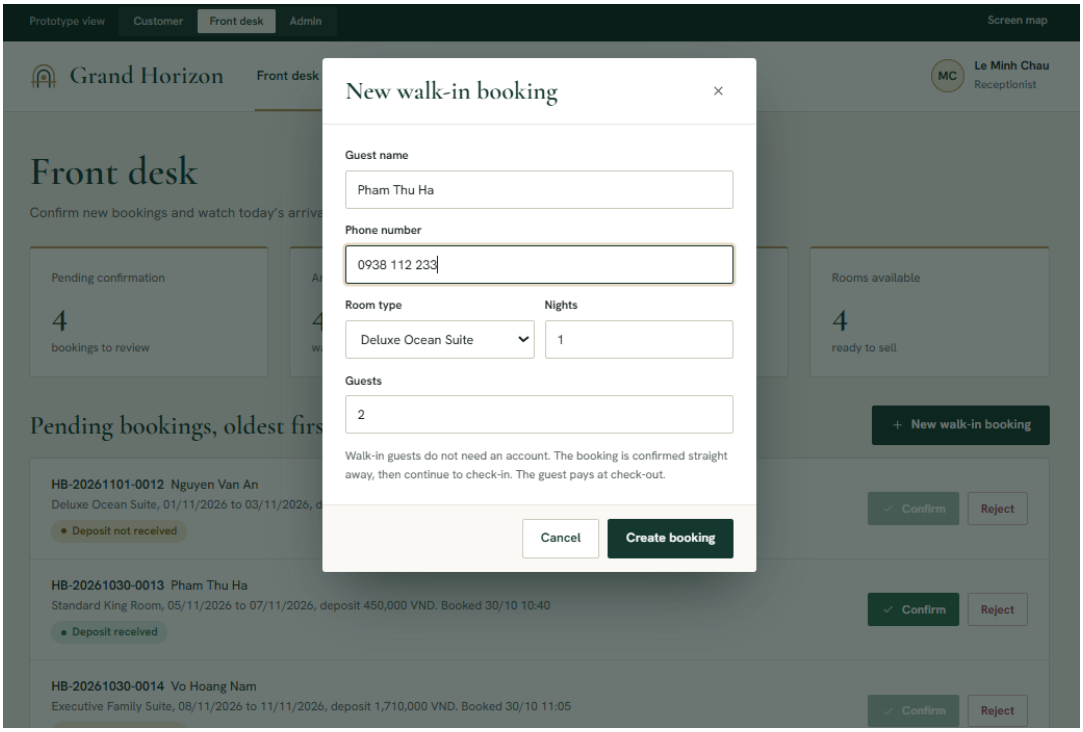


Figure 7.15: SCR-15 - Walk-in Booking Screen

Pop-up for creating a booking at the desk with Guest name, Phone number, Room type, Nights and Guests.

No customer account is needed: the booking is confirmed immediately, the guest continues to check-in and pays the full amount at check-out.

7.2.4 SCR-16: Check-in & Room Assignment

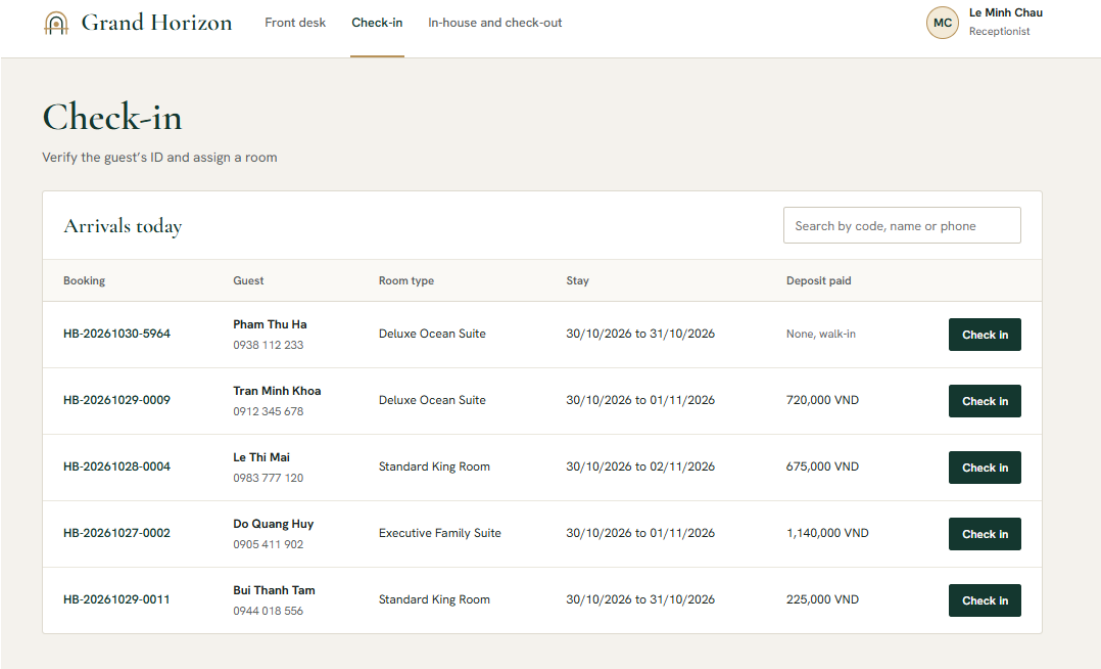


Figure 7.16: SCR-16 - Check-in: Arrivals List

Lists the guests arriving today with booking code, guest name and phone, room type, stay and deposit paid; the search box finds a booking by code, name or phone.

Each row has a Check in button that starts the identity check and the room assignment.

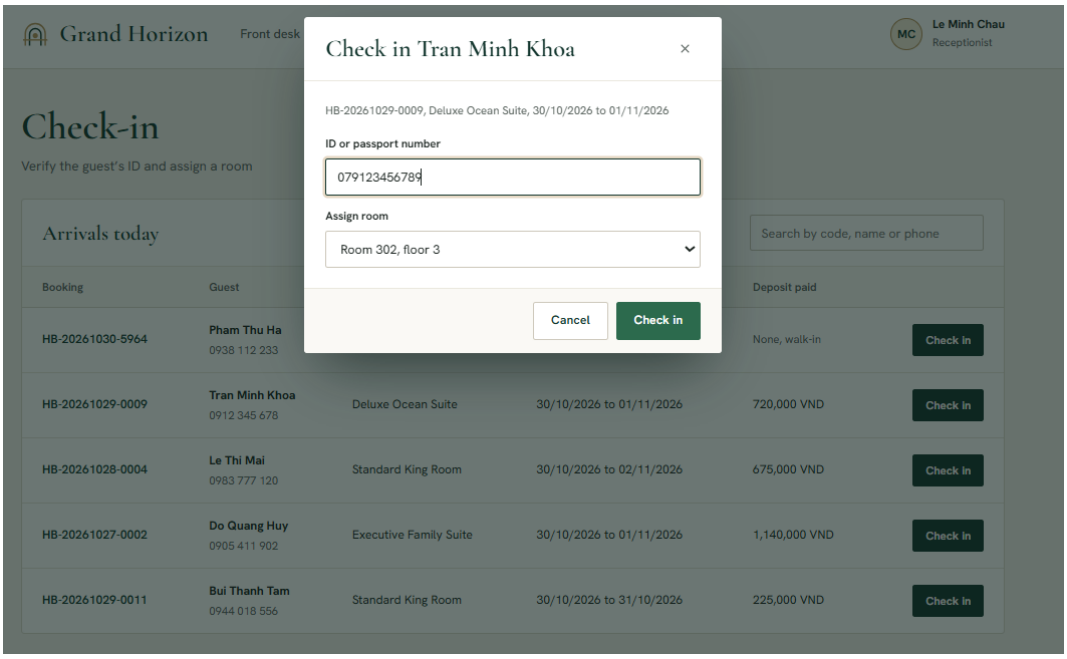


Figure 7.17: SCR-16 - Check-in: Room Assignment

The receptionist records the guest's ID or passport number and selects a room for the booked room type.

Only rooms of that type whose status is Available are offered; Check in completes the process and moves the guest to the in-house list.

7.2.5 SCR-17: Extra Services Screen

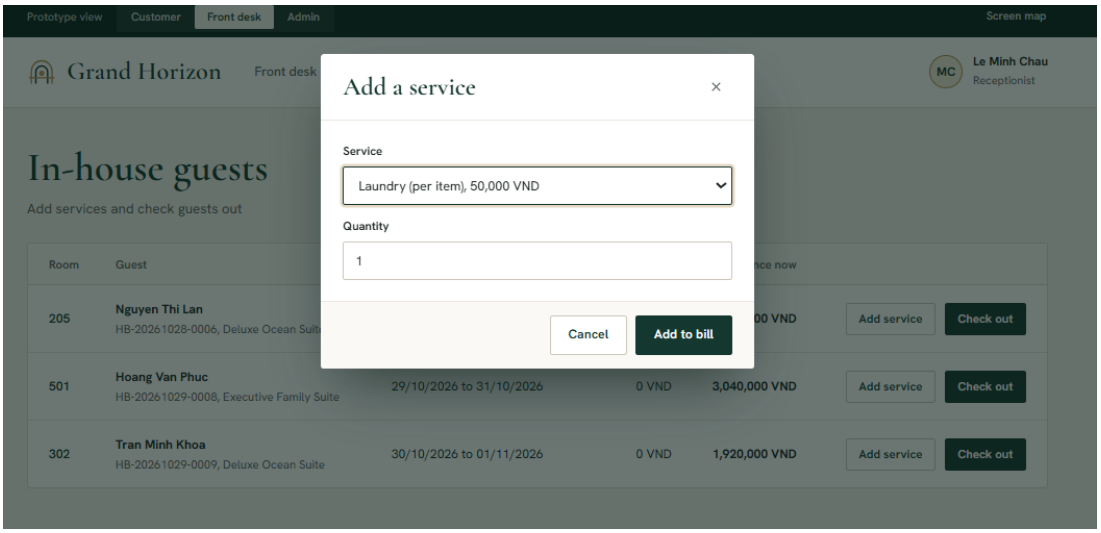


Figure 7.18: SCR-17 - Extra Services Screen

Pop-up opened from the In-house guests list to add a service to a guest's bill: the receptionist chooses the Service (shown with its unit price) and the Quantity.

Add to bill records the charge so that it appears on the guest's final invoice at check-out.

7.2.6 SCR-18: Check-out & Invoice Screen

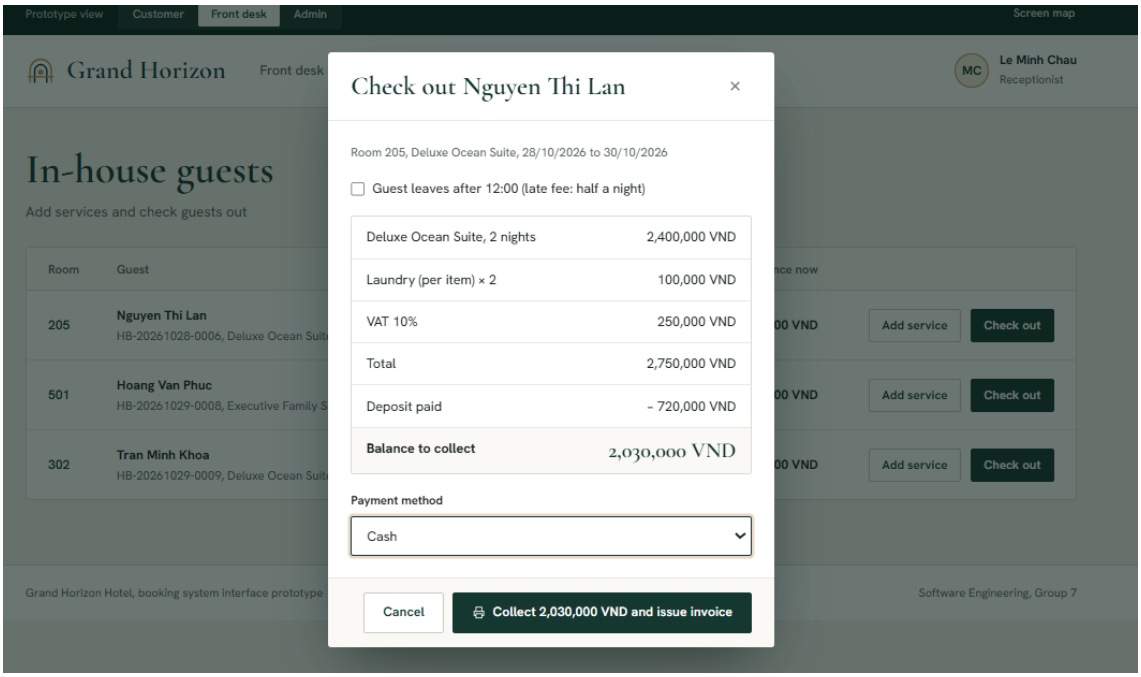


Figure 7.19: SCR-18 - Check-out and Invoice Screen

Check-out pop-up with the final bill: room charge, extra services, VAT 10%, total, deposit already paid and the balance to collect.

The receptionist can tick a late check-out (after 12:00, half a night is charged), choose the payment method and press Collect to complete the check-out and issue the invoice.

7.3 Admin Interfaces

7.3.1 SCR-19: Admin Dashboard (Reports)

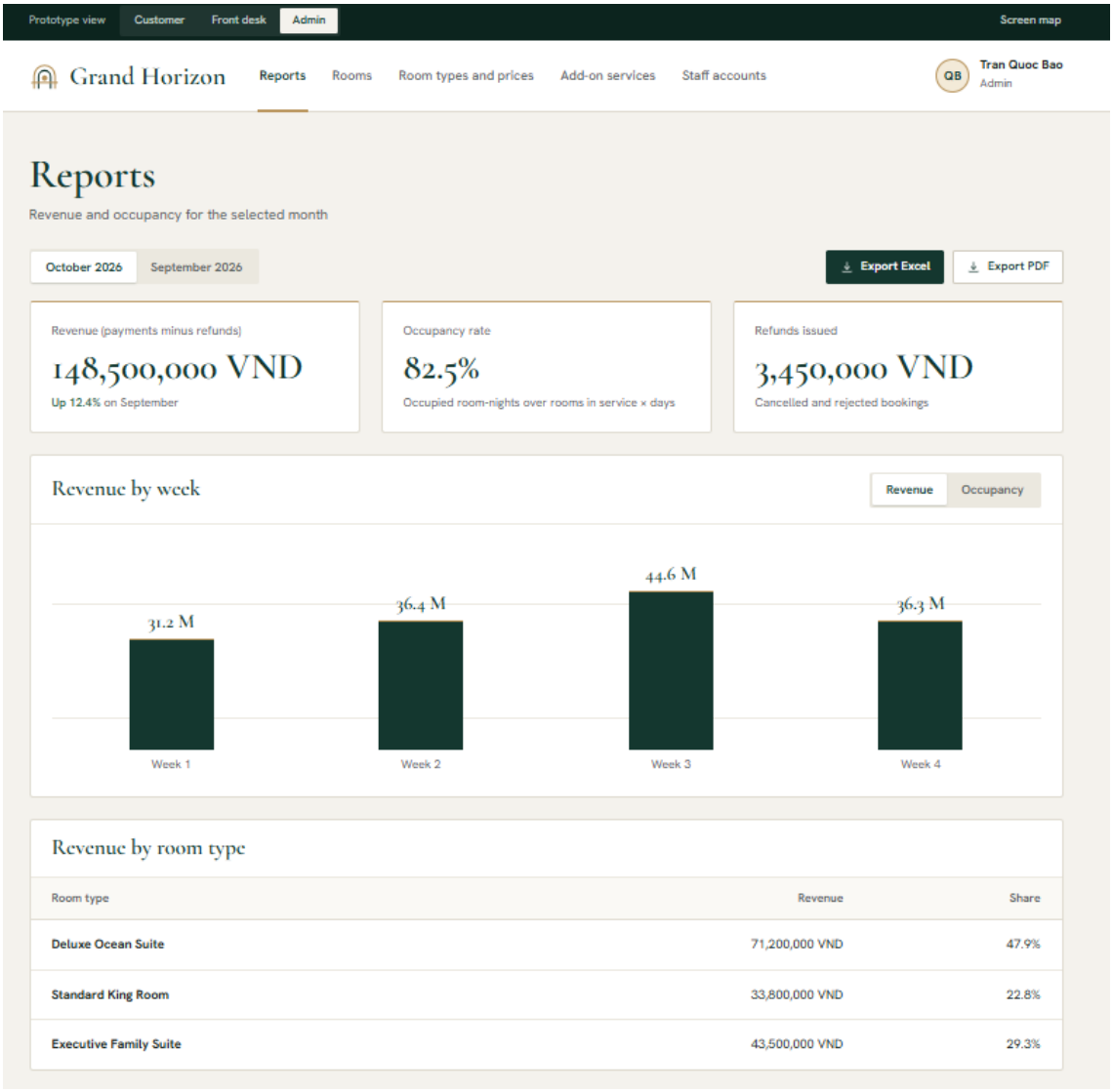


Figure 7.20: SCR-19 - Admin Dashboard (Reports)

Reports page for the selected month with Export Excel and Export PDF buttons and three indicators: revenue, occupancy rate and refunds issued.

Revenue is payments minus refunds and is compared with the previous month; occupancy is the occupied room-nights over the rooms in service multiplied by the days.

7.3.2 SCR-25: Revenue Report Screen

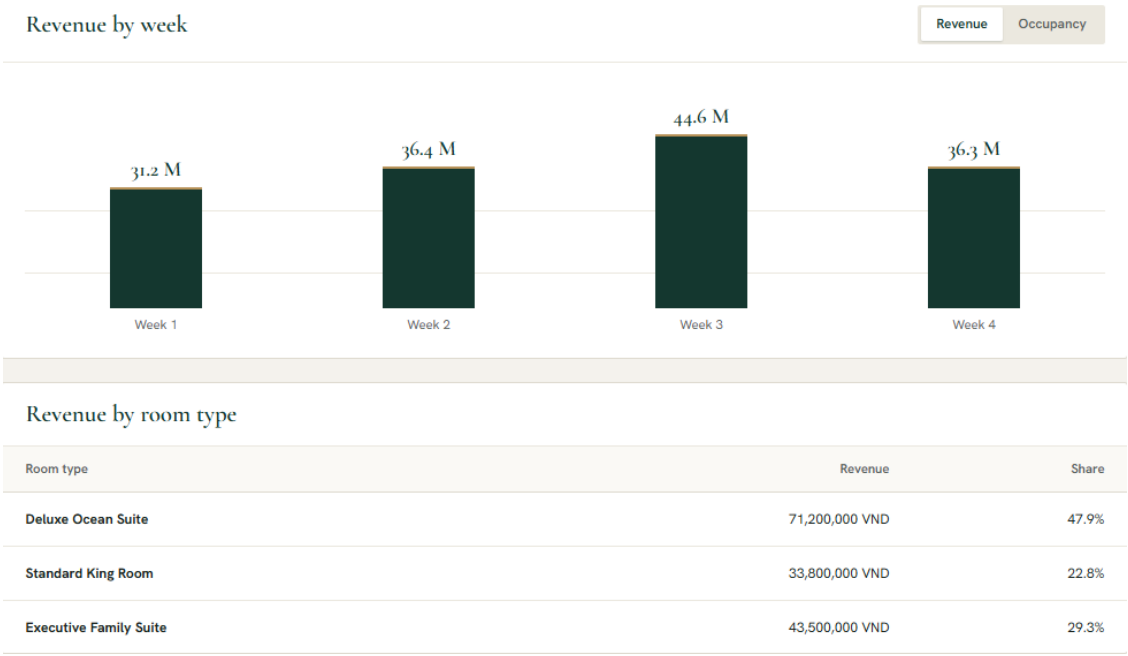


Figure 7.21: SCR-25 - Revenue Report Screen

The Revenue tab shows a bar chart of the revenue of each week in the selected month.

The table Revenue by room type lists the revenue and the share of every room type, so the manager can see which types earn the most.

7.3.3 SCR-26: Occupancy Report & Export

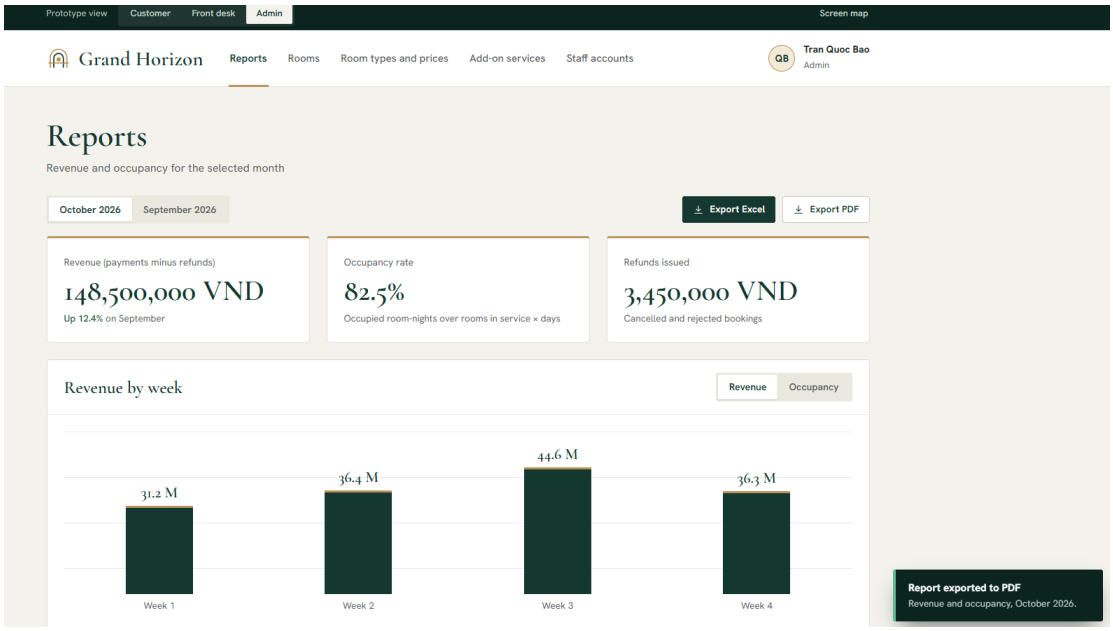


Figure 7.22: SCR-26 - Occupancy Report and Export

The Occupancy tab shows the occupancy rate of each week as a bar chart.

Export Excel and Export PDF produce the report of the selected month, and the system confirms the export with the message “Report exported to PDF”.

7.3.4 SCR-22: Room Management Screen

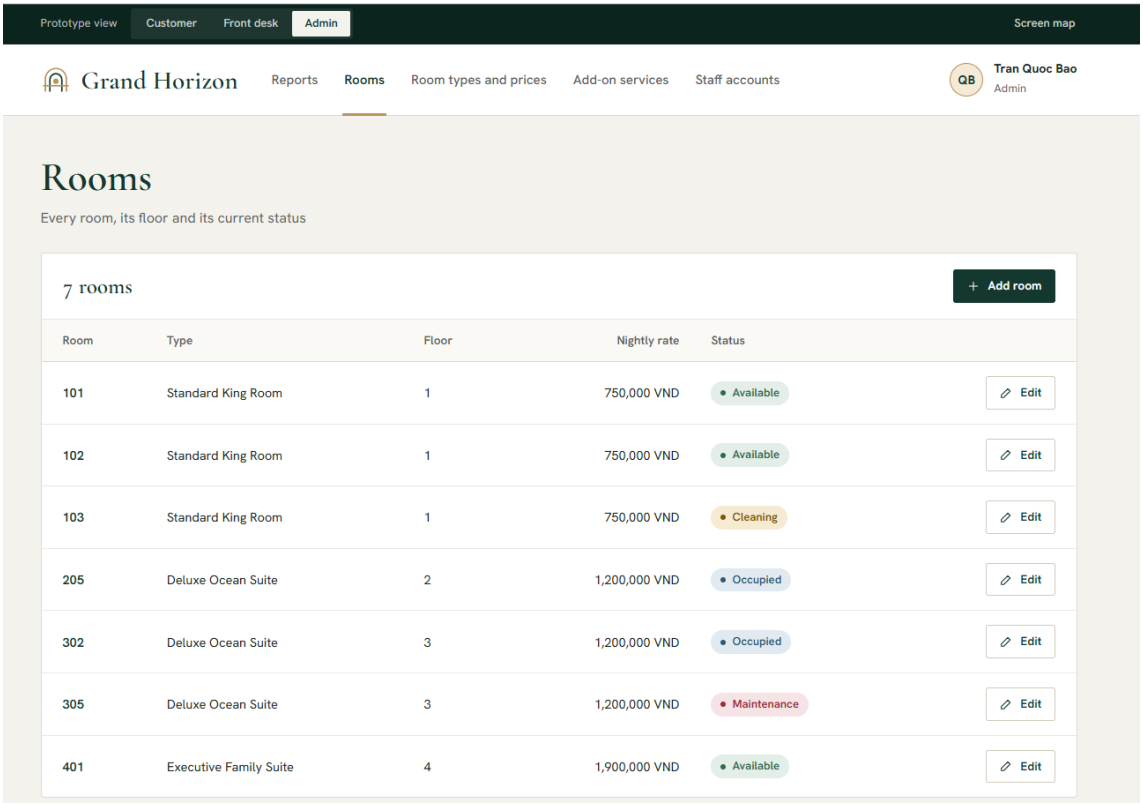


Figure 7.23: SCR-22 - Room Management Screen

Lists every room with room number, room type, floor, nightly rate and status (Available, Occupied, Cleaning or Maintenance).

The administrator adds a room with Add room and changes a room's details with Edit.

7.3.5 SCR-21 & SCR-23: Room Types & Price List

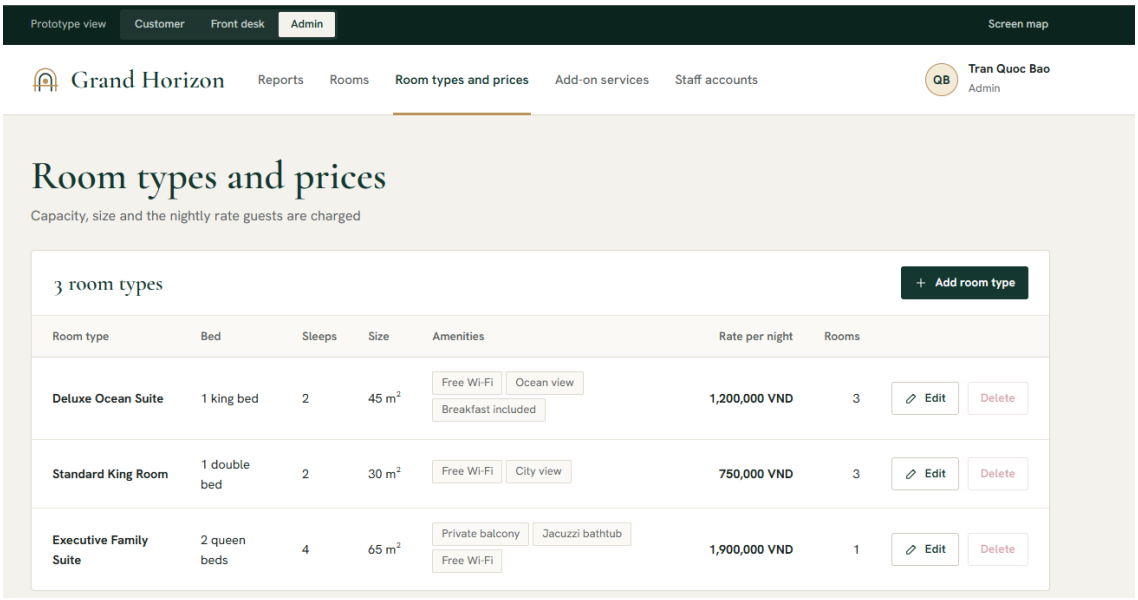


Figure 7.24: SCR-21 and SCR-23 - Room Types and Price List

Table of room types with bed, number of guests, size, amenities, rate per night and number of rooms, Add room type, Edit and Delete maintain the catalogue.

The nightly rate is maintained here, a new rate applies only to new bookings, and bookings already made keep the price they were booked at.

7.3.6 SCR-24: Add-on Services Screen

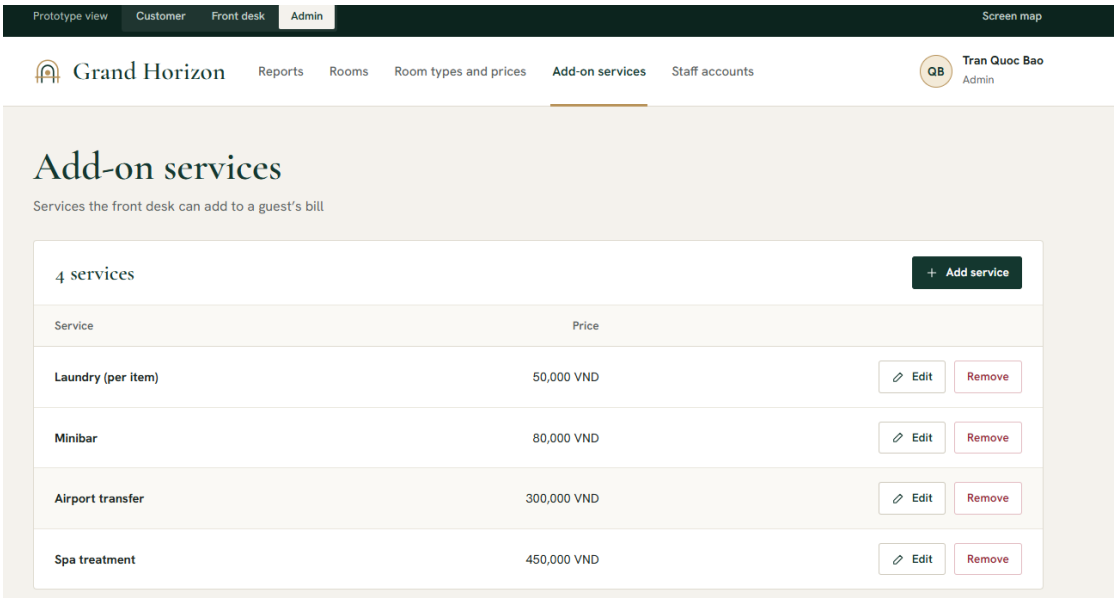


Figure 7.25: SCR-24 - Add-on Services Screen

Lists the add-on services the front desk can charge (Laundry, Minibar, Airport transfer and Spa treatment) with their prices.

The administrator can add, edit or remove services; a price change does not affect services that are already on a guest's bill.

7.3.7 SCR-20: Staff Accounts Screen

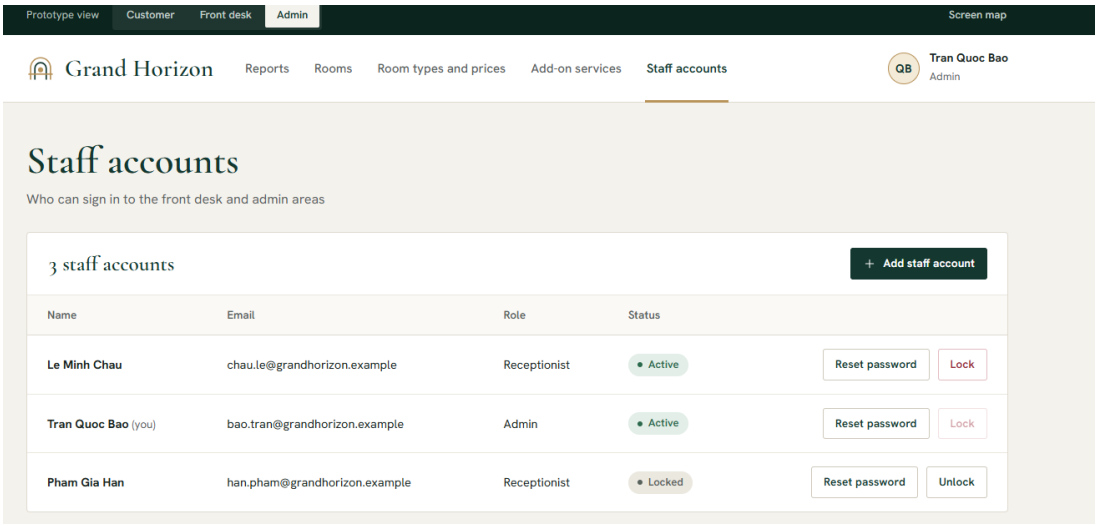


Figure 7.26: SCR-20 - Staff Accounts Screen

Lists the staff accounts with name, email, role (Receptionist or Admin) and status (Active or Locked), add staff account creates a new one.

The administrator can Reset password and Lock or Unlock an account, the signed-in administrator cannot lock their own account.